Hurricanes analysis with Holland models

BY

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Chapter 1

Hurricane Description

1.1 Introduction & Terminology

Tropical cyclones are strong and destructive weather phenomena, characterised by a system of thunderstorms that are rotating around the low-pressure centre and generating strong winds, rainfalls, and storm surges [1]. These atmospheric vortices are forming above the warm tropical and subtropical ocean waters, harvesting energy from the heat generated from the condensation of water vapour [1]. Although tropical cyclones are appearing in multiple oceanic basins all over the world, they are named differently depending on the region: hurricanes in the North Atlantic and East Pacific oceans, typhoons in the West Pacific Ocean, and just tropical cyclones in the South Pacific and Indian oceans [2]. The terminology reflects historical and regional conventions; there are no fundamental differences in the physical nature of these storms. This project is focused specifically on hurricanes of the North Atlantic – tropical cyclones that form in the Atlantic Ocean, Caribbean Sea, and Gulf of Mexico. The basin of the North Atlantic is an especially interesting region for research for multiple reasons. Firstly, it has one of the longest and most informative records across all the basins of tropical cyclones with systematic documentation that goes almost a century back with detailed satellite images since the 1960s [3]. Secondly, hurricanes of the North Atlantic have social and economic impacts which heavily affect populated areas from the islands of the Caribbean Sea up to the eastern parts of the United States of America and sometimes even parts of the Canadian Atlantic coast [4]. Finally, the basin demonstrates significant change in activity of hurricanes in multiple terms of time – from separate storms, which can intensify or weaken, to seasonal variations in storm frequency, to decadal cyclones in levels of activity [5]. These behaviours make the North Atlantic a perfect place for testing mathematical models of structures and the intensity of hurricanes.

This chapter provides fundamental knowledge necessary for understanding hurricanes and the context for mathematical modelling approaches, such as the Holland model, which is a parametric model that this report will focus on.

1.2 Physical Structure and Anatomy

Hurricanes have a well-organised three-dimensional structure with distinctive features, which differentiate them from other atmospheric phenomena. Understanding of this structure is critically important for modelling of wind distribution and intensity of the hurricane. In this section we will explore key components of a hurricane and describe typical sizes and characteristics of these elements.

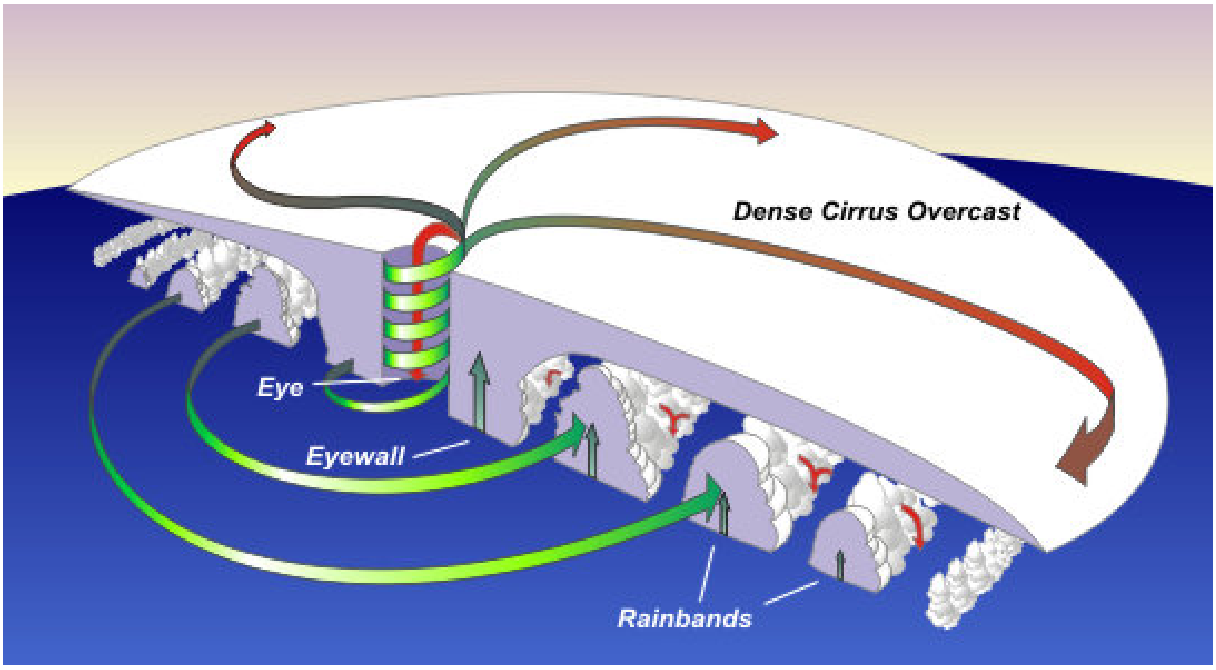


Figure 1.1: Hurricane cross section from NOAA article "Tropical Cycle Structure"[6]

1.2.1 The Eye

The eye is the central area of the hurricane. It can be observed in Figure 1.1. It is characterized by being relatively calm with a clear sky or broken clouds and having the lowest atmospheric pressure in the whole system. The diameter of the eye usually ranges from 20 to 60 kilometers [6]. Winds inside the eye are relatively weak, usually less than 25 km/h, and the air descends (downward movement). The temperature in the eye is higher than in the surrounding storm [6]. The forming of the clearly defined eye is a sign of a mature hurricane. The size and the form of the eye can change during the life cycle of the hurricane [6].

1.2.2 The Eyewall

The eyewall is a convection ring made of high thunderstorm clouds surrounding the eye, where the strongest winds and the most intense rainfalls of the hurricane are observed [6]. With approximately a 119 km/h sustained rotation of air around the cyclone, it balances the flow to the centre, forcing the air to elevate around 16–32 kilometers from the centre, creating the eyewall. Changes in the structure of the eyes and eyewall may change the wind speeds, which is an indicator of the intensity of the hurricane [6]. Eyes may expand and shrink and double (concentric) eyewalls may form [6].

1.2.3 Rainbands

Curved bands of clouds and thunderstorms that spiral away from the eyewall are called rainbands [6]. For visualization watch Figure 1.1. They are represented by precipitation in the form of showers and sometimes thunderstorms, which spread radially from the centre of the tropical cyclone [2]. These bands may produce abundant rains and strong wind gusts; however, they are much weaker than in the eyewall [6]. Rainbands may stretch out for hundreds of miles from the centre of the hurricane and often are the first sign of an incoming storm, arriving to the coast much earlier than the centre of the hurricane. Structure and intensity of rainbands may change as the hurricane develops. In some intense tropical cyclones, external rainbands can organise into an outer ring of thunderclouds, which can lead to the forming of a new eyewall and starting the eyewall replacement cycle [6]. The new eyewall slowly moves inward and takes away the necessary moisture and momentum from the inner eyewall. During this phase the tropical cyclone is weakening. Ultimately, the outer eyewall completely replaces the inner eyewall, and the storm may have the same intensity as before or, in some cases, even be stronger.

1.3 Metrics of intensity

There are multiple metrics that are used to characterize the strength of tropical cyclones. In this section, we will focus on parameters that are important for parametric modelling.

1.3.1 Central pressure

Minimum central pressure is a fundamental metric of the intensity of the hurricane. Central pressure is measured in the centre of the hurricane, where it is the lowest in the whole structure [2]. There is an empirical connection between central pressure and maximum wind speeds: lower central pressure usually corresponds to stronger winds. However, this connection is not universal and depends on multiple factors such as the size of the storm, width, movement speed, and pressure of the environment [7].

1.3.2 Radius of Maximum Winds (RMW)

The Radius of Maximum Winds, or RMW for short, is a distance from the center of the hurricane to the location of the maximum winds [2]. RMW is a central parameter in the parametric wind models, such as the Holland model. It is critically important for modelling wind distribution, as it determines the radial wind gradient: lower RMW corresponds to a steeper gradient and more concentrated strong winds [8].

The next chapter explores the integration of these metrics into a parametric model.

Chapter 2

Holland Model

2.1 Model description

The parametric approach of Holland to modeling radial wind profiles in hurricanes is a mathematical method that describes how the wind speed changes with distance from the center of the storm.

The key idea of the model is to represent the hurricane wind field using a small number of physically meaningful parameters, such as central pressure, ambient pressure, maximum wind speed, and the radius of maximum winds. These quantities are typically available from hurricane databases, making the model practical for both research and engineering applications. This model is used in the reconstruction of historical wind fields, estimation of extreme hazards, and development of engineering design standards [9]. In this research, we will look at four different hurricanes from 2024 and try to get as accurate an output as possible.

We will be looking at the original Holland's model of 1980 [9] and revised Holland's model of 2010 [8] and refer to them as H80 and H10, respectively, for convenience.

2.1.1 H80 Model (Holland 1980)

The original Holland model describes the radial structure of a cyclone through analytical expressions for pressure and wind speed.

Pressure equations:

$$P_c = P_n - \Delta P e^{-(r_{vm}/r)^b} \tag{2.1}$$

Where parameter b is,

$$b = \frac{V_m^2 \rho e}{\Delta P} \tag{2.2}$$

Where P_c – pressure on radius on specific radius r, P_n – ambient (environmental) pressure, ΔP – pressure difference between P_n and P_c, ρ – air density at gradient level (kg m^{-3}), V_m – the maximum wind at the gradient level (in m/s). Ambient pressure is usually assumed to be equal to standard atmospheric pressure at sea level (1013 mb), although it may vary [10].

This parameter controls the shape of the storm. The larger it is, the more compact and sharply peaked the storm will be. Smaller b leads to a broader and more gradual wind profile.

Wind Profile

The wind speed can be expressed as:

$$V(r) = V_m \left\{ \left(\frac{r_{V_m}}{r} \right)^b e^{1-(r_{V_m}/r)^b} \right\}^{0.5} \tag{2.3}$$

Where r_{V_m} – radius of maximum wind, r - radius at which the wind speed being calculated, V_m – the maximum wind at the gradient level (in m/s), b - scaling parameter b that was mentioned above.

This expression describes how the wind speed increases toward the radius of maximum wind and then decays outward.

Limitation of H80

A key limitation of the H80 model is that it is formulated in terms of gradient-level wind [8], which is not directly observed in standard hurricane datasets such as HURDAT2 [11]. As a result, additional assumptions are required to convert between surface and gradient wind speeds, which may lead to more errors and inaccuracies.

2.1.2 H10 Model (Holland 2010)

The revised model (H10) extends the original formulation to operate directly with surface wind observations, which are available from aircraft reconnaissance.

The main improvement is the introduction of a variable shape parameter $x(r)$, which allows greater flexibility in matching observed wind profiles.

Wind Profile

The surface wind is given by:

$$V(r) = V_m \left\{ \left(\frac{r_{V_m}}{r} \right)^{b_s} e^{1-(r_{V_m}/r)^{b_s}} \right\}^x \tag{2.4}$$

Where r_{V_m} – radius of maximum winds on surface level, V_m – the maximum wind at the surface level, and b_s – is a scaling parameter that defines the proportion of the surface pressure near the maximum wind radius, x – is a variable exponent controlling the profile shape.

Parameter b_s

$$b_s = \frac{V_m^2 \rho e}{\Delta P} \tag{2.5}$$

It is similar to H80 model but applied parameters on surface level.

Variable parameter $x(r)$

A key innovation of the H10 model is allowing the exponent x to vary with radius:

$$x(r) = 0.5 + (r - r_{V_m}) \left(\frac{x_n - 0.5}{r_n - r_{V_m}} \right) \tag{2.6}$$

With x_n being the adjusted exponent to fit the peripheral observations at radius r_n.

This represents a linear interpolation between $x = 0.5$, which is the constant over the whole profile in H80 model case, at the radius of maximum wind, which is assumed by the model inside the radius of maximum wind and consistent with the original H80 model, and $x = x_n$ at a known outer radius r_n. The named parameter allows the model to incorporate observed wind speeds at multiple radii, better reproduce the outer wind structure, and improve agreement with real hurricane wind fields.

2.2 H80 & H10 Wind Profiles

In this section, the Holland (1980) and Holland (2010) models are applied to a selection of hurricanes at their peak intensity in order to compare their resulting wind profiles. There are two main reasons for focusing at the peak intensity. First, hurricanes at their peak typically have the most complete observational records, including all the parameters required for the application of the Holland model. Second, from a hazard assessment perspective, it is most appropriate to evaluate the worst-case scenario, when a hurricane has the greatest potential to cause damage to infrastructure and affected populations within the impacted region.

The data used for this analysis are obtained from the National Hurricane Center HURDAT2 database [11], which contains historical records of tropical cyclones from 1851 to the present. In this study, only recent hurricanes are considered, as these provide the most complete datasets. In particular, we will look at four hurricanes from 2024, such as Oscar, Helene, Beryl, and Kirk. These hurricanes were the strongest in the year of 2024 in the North Atlantic and had all the necessary observational records available from HURDAT2 [11].

All computations and visualisations were performed using the Python programming language in the JupyterLab environment. Relevant code implementations are provided in Appendix A.

2.2.1 Data preparation

For each hurricane, the following parameters were extracted from the HURDAT2 [11]: minimum central pressure P_c, maximum wind speed V_m, radius of maximum wind r_{v_m}, and wind radii corresponding to 34, 50, and 64 knots wind speeds. As well as the date, time, longitude, and latitude of the centre of the hurricane (see Appendix A, pp. 42,43).

The wind radii are provided in four directions from the centre of the hurricane: North-West, North-East, South-East, and South-West, resulting in twelve directional measurements. At first, we will follow the steps of the original model approach and model the wind profile in one plane. Hence, we will take the median from four directional wind measurements. Median was chosen instead of mean for the directional observational data for computing original H10 and H80 models because the difference between quadrants in some cases is too high, and the mean would be highly affected by the anomalies that were discovered later and affected the wind profiles in the process. So it was decided to use the median instead.

2.2.2 Model implementation

The H80 was implemented in Python (see Appendix A, pp. 45) using equations (2.2) and (2.3), producing the gradient-level wind profile, which translates to surface-level wind with an approximate 20% loss in speed [12].

The H10 model was implemented in Python (see Appendix A, pp. 45, 46) using equations (2.4) - (2.6), which allow the use of surface wind observations and introduce a variable parameter $x(r)$.

Both models were implemented with constant values of ambient (environmental) pressure $P_n =$ 1013.5 hPa and air density $\rho = 1.15$ kg m^{-3} [10], as these variables are not available in HURDAT2.

Since continuous wind observations at every radius are not available, the H10 model is constrained using the known wind radii 34, 50 and 64 knots. These discrete data points are used to construct a piecewise approximation of the parameter $x(r)$.

2.2.3 Construction of the $x(r)$ profile

The parameter $x(r)$ is defined in implemented H10 model as follows:

1. For $r \leq r_{v_m}, x = 0.5$,
2. For $r_{v_m} < r \leq r_{64}, x(r)$ is linearly interpolated between 0.5 and x_{64},
3. For $r_{64} < r \leq r_{50}, x(r)$ is linearly interpolated between x_{64} and x_{50}),
4. For $r_{50} < r \leq r_{34}, x(r)$ is linearly interpolated between x_{50} and x_{34}),
5. For $r > r_{34}, x = x_{34}$ is assumed constant.

Where r_{34}, r_{50}, r_{64} are radii of 34, 50, and 64 knots wind speed, respectively, and x_{34} is the calculated parameter x at 34 knots wind speed radius.

This approach ensures that the model matches the observed wind speeds at the known radii while maintaining a continuous radial profile. Currently, this is the only wind measurement available over the Atlantic Ocean, as NOAA was defunded by its government and limited in the availability of more aircraft reconnaissance data, as it is provided directly by the U.S. Air Force.

2.2.4 Analysis and Results

Graphical comparison and quantitative analysis were performed using Python (see Appendix A, pp. 50, 51). The analysis is based on differences in wind speed at each radial distance, with the root mean square difference (RMSD) calculated over defined radial intervals. These intervals represent key regions of the hurricane structure, including the inner core, from the centre to the radius of maximum wind, and successive bands extending 50–100 km beyond the eyewall. For each interval, mean wind speeds for both the H80 and H10 profiles, maximum deviations, RMSD, and normalised RMSD (%) were computed to quantify the magnitude of inter-model differences relative to the wind field intensity.

Hurricane Oscar

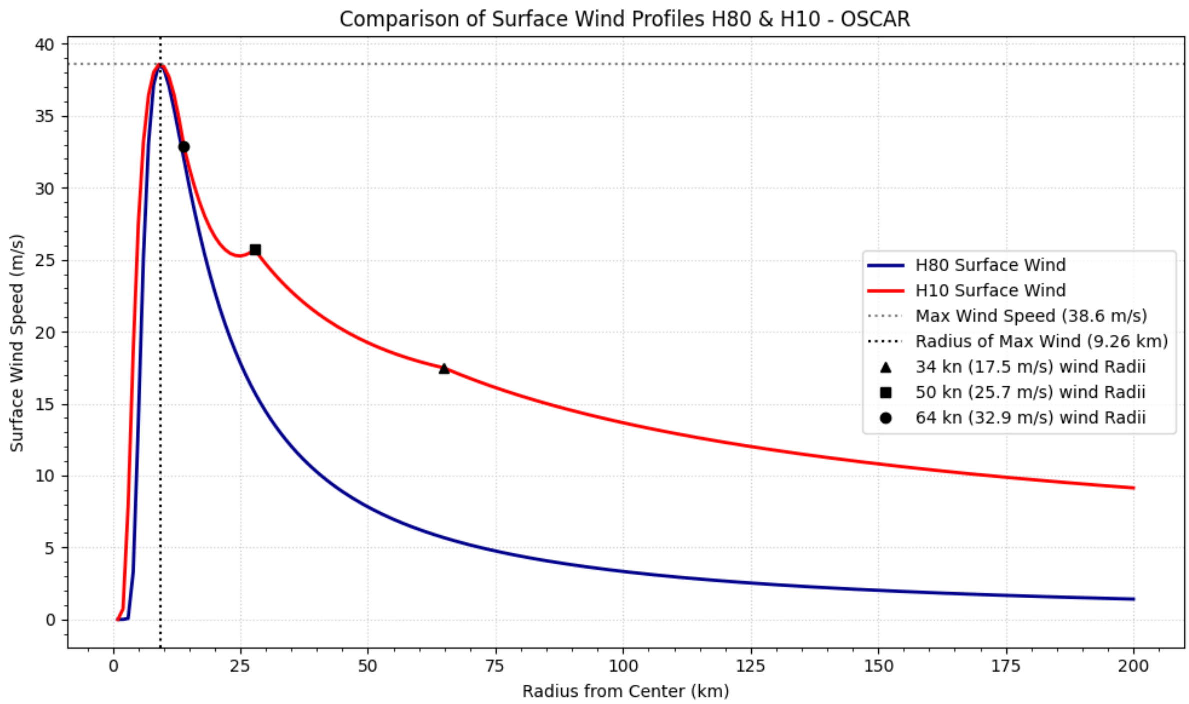


Figure 2.1: H80 & H10 Wind Profiles of Hurricane OSCAR at its peak intensity in 2024

Quantitative analysis H80 & H10 - OSCAR					
Interval (km)	H10 Mean (m/s)	H80 Mean (m/s)	RMSD (m/s)	Normalised RMSD (%)	Maximum Deviation (m/s)
0 - 5	6.8	0.84	8.61	225.22	25.25
5 - 55	25.66	17.89	8.84	40.59	22.97
55 - 105	15.74	4.63	11.11	109.08	6.77
105 - 200	10.80	2.04	8.78	136.71	10.42

As we can see on the Figure 2.1 for Hurricane Oscar, both the H80 and H10 models capture a similar peak wind speed near the storm centre. However, their radial decay differs significantly. The H10 model produces a broader wind field, maintaining higher wind speeds at larger radii, where the H80 model shows a more rapid decline beyond the radius of maximum wind. Compared with the observed wind radii 34, 50, and 64 knots, represented by triangular, square, and circular markers respectively, the H10 profile generally aligns more closely with the outer wind structure, while H80 underestimates the wind speeds at longer distances. Quantitatively, this is reflected in substantial RMSD values across radial intervals, with particularly large deviations near eyewall and beyond, with H10 profile showing above 100% increases in wind speeds.

Hurricane Helene

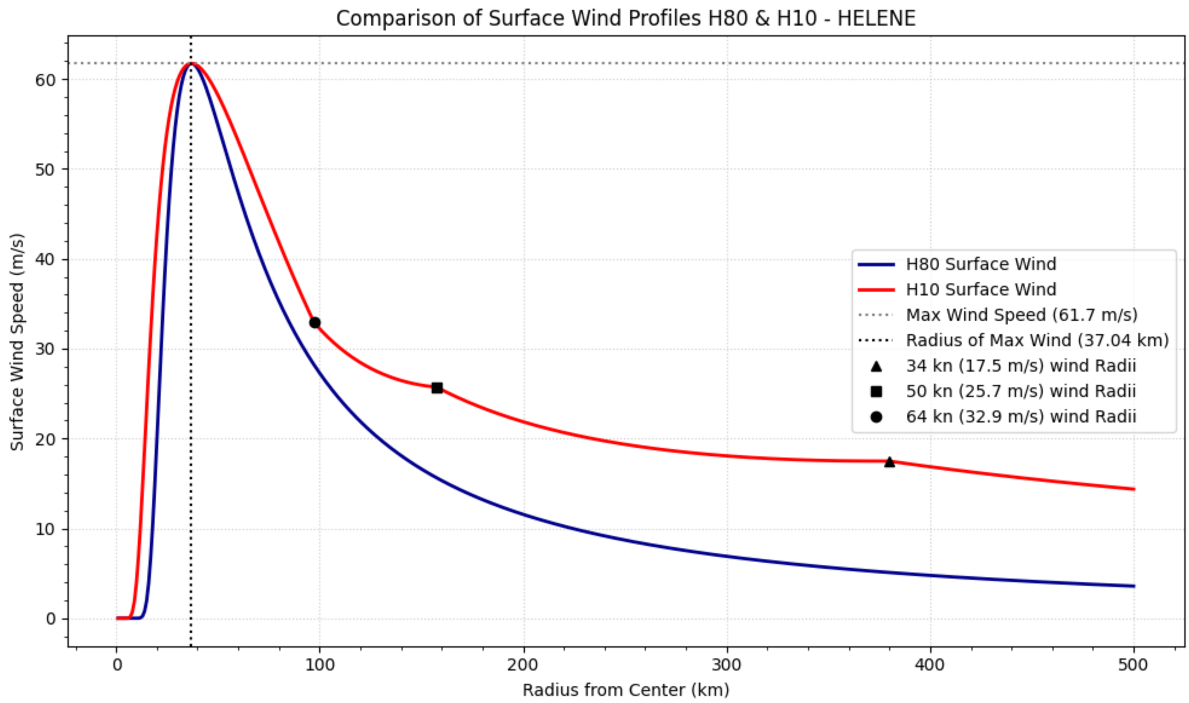


Figure 2.2: H80 & H10 Wind Profiles of Hurricane HELENE at its peak intensity in 2024

Quantitative analysis H80 & H10 - HELENE					
Interval (km)	H10 Mean (m/s)	H80 Mean (m/s)	RMSD (m/s)	Normalised RMSD (%)	Maximum Deviation (m/s)
0 - 20	11.35	2.05	13.77	205.36	25.25
20 - 70	56.04	50.77	7.41	13.87	22.97
70 - 120	35.82	29.94	5.91	17.98	6.77
120 - 220	24.33	14.85	9.55	48.77	10.42
220 - 320	18.9	8.03	10.87	80.71	11.42
320 - 500	16.39	4.72	11.67	110.53	12.39

For Hurricane Helene, on Figure 2.2, H10 model maintains significantly higher wind speeds with a more gradual radial decay, resulting in a broader wind field, where the H80 model shows a much steeper decline beyond the radius of maximum wind. Comparison with the observed wind radii 34, 50, and 64 knots, represented by triangular, square, and circular markers respectively, indicates that the H10 aligns more closely with the extent of the outer wind structure, while the H80 increasingly underestimates wind speeds at larger radii. This is supported by the quantitative analysis, where normalised RMSD values remain elevated in the outer intervals, exceeding 80%, and maximum deviations remain above 10 m/s after the 100 kilometer mark. Overall, H10 provides a more realistic representation of the spatial extent of the wind field, whereas H80 produces a more compact profile that does not capture the observed wind speeds.

Hurricane Beryl

Quantitative analysis H80 & H10 - BERYL					
Interval (km)	H10 Mean (m/s)	H80 Mean (m/s)	RMSD (m/s)	Normalised RMSD (%)	Maximum Deviation (m/s)
0 - 10	5.39	0.18	9.66	345.98	24.08
10 - 60	52.68	40.63	13.62	29.19	29.88
60 - 110	26.37	10.37	16.02	87.22	17.35
110 - 210	18.8	3.63	15.18	135.3	16.51
210 - 350	13.54	1.53	12.03	159.66	13.72

On Figure 2.3, we can see similar picture to the previous two hurricanes with H80 wind speed values are decaying rapidly while H10 is significantly with higher values after 25-30 kilometer

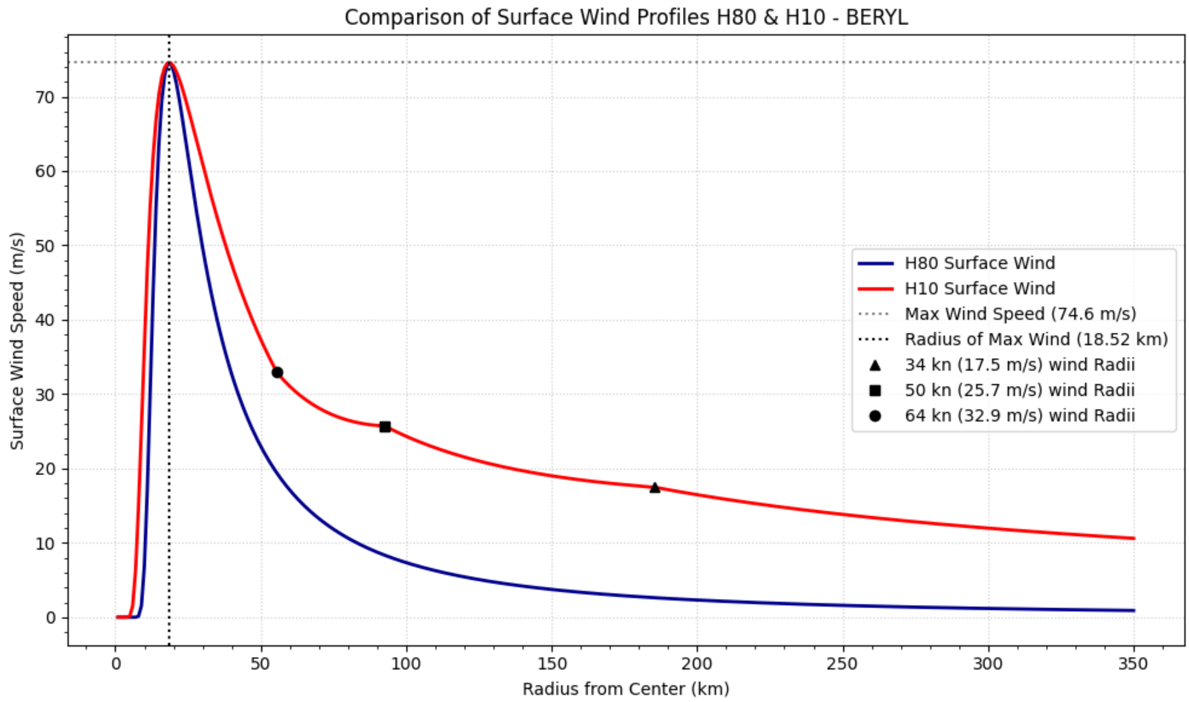


Figure 2.3: H80 & H10 Wind Profiles of Hurricane BERYL at its peak intensity in 2024

mark.

Hurricane Kirk

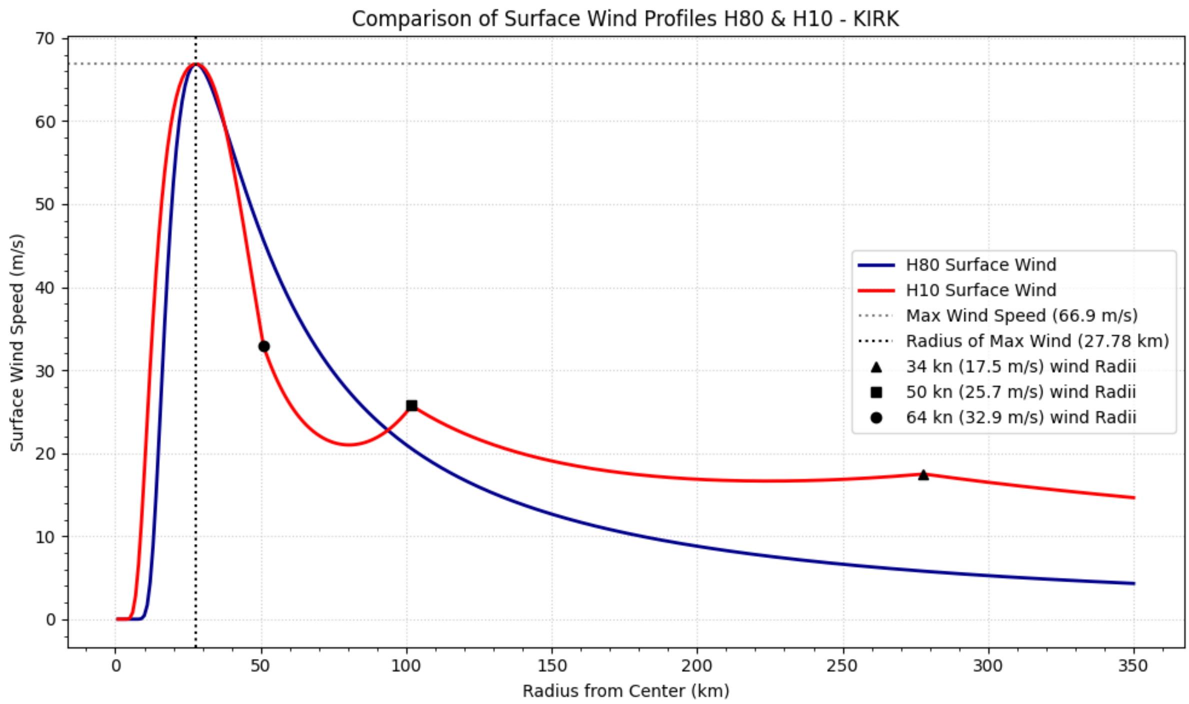


Figure 2.4: H80 & H10 Wind Profiles of Hurricane KIRK at its peak intensity in 2024

Quantitative analysis H80 & H10 - KIRK					
Interval (km)	H10 Mean (m/s)	H80 Mean (m/s)	RMSD (m/s)	Normalised RMSD (%)	Maximum Deviation (m/s)
0 - 15	12.28	2.16	14.94	206.87	27.37
15 - 65	48.73	51.21	9.65	19.31	24.76
65 - 115	22.8	24.88	6.34	26.59	11.46
115 - 215	18.69	11.78	6.97	45.73	8.61
215 - 350	16.80	6.26	10.58	91.74	11.7

Figure 2.4 presents the wind profile comparison for Hurricane Kirk, which exhibits a slightly different pattern compared to the previous cases. Beyond the radius of maximum wind between approximately 40 and 100 km, the H10 model shows a pronounced reduction in wind speed, resulting in a local dip not observed in the H80 profile, which maintains a smoother decay. Beyond the observed 50 knot wind radius, both models begin to follow a more similar trend, where the H10 again produces higher wind speeds at larger radii. This is supported by the quantitative analysis, where normalised RMSD values indicate differences of up to 90% in the outer intervals. Overall, while H10 generally represents a broader wind field, its non-monotonic behaviour in the mid-radii distinguishes it from the smoother and more consistently decaying H80 profile.

Results

Across the majority of cases (OSCAR, HELENE, and BERYL), the H10 model produces wind profile that closely aligns with the observed wind speeds. In particular, the model successfully reproduces the wind speeds at 34, 50, and 64 knots, demonstrating the advantage of incorporating additional observational constraints through the parameter $x(r)$.

In contrast, the H80 model exhibits a much more rapid decay of wind speed with radius. This reflects its more restrictive structure, as it relies on a constant shape parameter and does not incorporate additional wind observations beyond the maximum wind.

These results confirm that the H80 model tends to underestimate wind speeds compared to H10, particularly in the inner and intermediate regions of the storm. This behavior is consistent with the structural limitations of the H80 model, which relies on a fixed profile shape and does not incorporate additional observational constraints from wind radii.

A similar pattern was observed for the other hurricanes considered in this study, reinforcing the conclusion that the H10 model provides a more flexible and realistic representation of the radial wind structure.

However, the case of Hurricane KIRK reveals a limitation of the H10 approach. In this instance, the model shows a noticeable local dip from the observed wind radius at 64 knots, resulting in a non-smooth behaviour of the wind profile. This suggests that the interpolation of $x(r)$ may become unstable when the input data are inconsistent or when the spacing between wind radii is irregular. Hence, H10 model, even though superior to the H80 model, is also far from perfect.

Several factors may contribute to these discrepancies. First, the assumption of a constant ambient (environmental) pressure P_n may not accurately represent the surrounding environment of the hurricane. Second, the median value of the recorded wind radii idealizes the hurricane as a perfect circle, which leads to missing the asymmetry of the real hurricanes.

Chapter 3

Research of Possible Solutions

3.1 Finding Environmental Pressure

3.1.1 Use of Reanalysis Data and Satellite Imagery

To address the limitation of assuming a constant ambient pressure P_n in the previous models' implementations, reanalysis data from the ERA5 dataset were introduced. Provided by the Copernicus Climate Data Store [13], ERA5 offers hourly estimates of atmospheric variables on a global grid with high spatial and temporal resolution.

In this study, surface pressure fields were extracted from ERA5 at the exact time of peak intensity of each hurricane (see Appendix B, pp. 56, 57). These fields provide a spatial representation of the ambient pressure around the centre of the hurricane, allowing for a more realistic interpretation of the pressure distribution compared to the constant-pressure assumption used in earlier modelling stages.

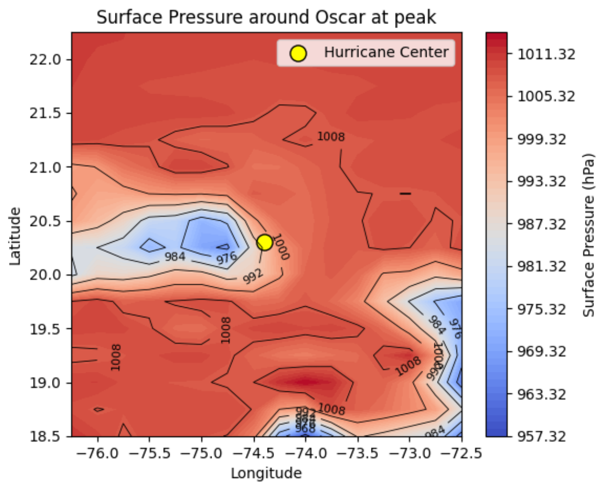
To support the physical interpretation of these pressure fields, satellite imagery from the GOES-16 satellite, Band 13 in particular, was also used [14]. Where Band 13 corresponds to the infrared window channel, which is commonly used to observe cloud-top temperatures and cyclone structures. This allows us to have a direct visual comparison between the pressure field and the observed density of the hurricane system. To put it simply, surface pressure in the storm is at its lowest closer to the eye of the storm, and the darker image on GOES16 imagery means cold temperatures and intense thunderstorm systems with transition to brighter colors with less intense but still storm structures.

3.1.2 Pressure Field Analysis

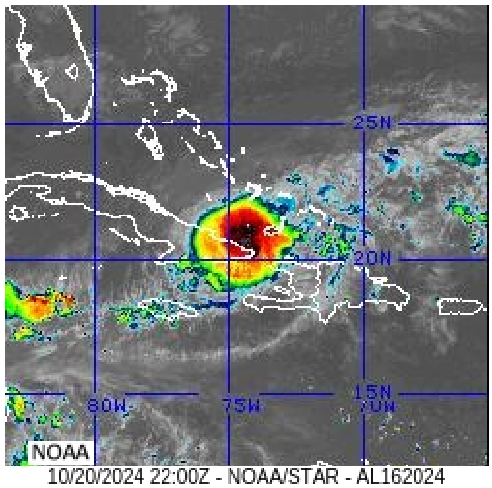
Hurricane Oscar at peak

Figure 3.1 shows the spatial distribution of surface pressure around hurricane OSCAR alongside the corresponding satellite image. The pressure field does not present a clear minimum at the storm centre, which can be explained by the geographical position of the hurricane at that moment. As it is known that the pressure tends to drop with surface elevation [12].

The satellite image confirms the presence of a well-defined cyclone structure, although some asymmetry is visible in the cloud field. This supports the observation that real hurricanes deviate from the idealised symmetric structure assumed in the Holland models.



(a) Surface Pressure Field



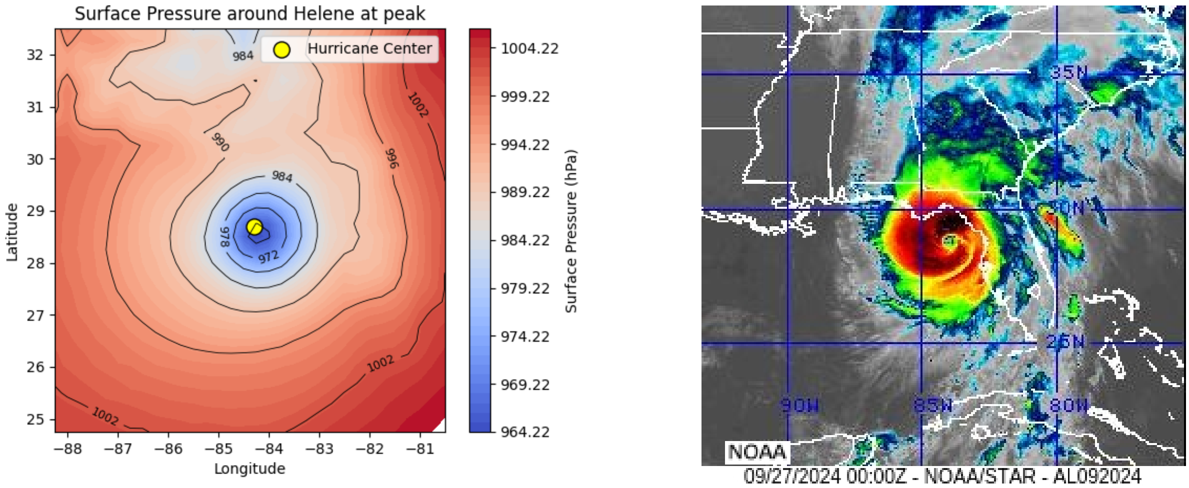
(b) GOES-16 Band 13 Image

Figure 3.1: Comparison of Real Surface Pressure data and the Satellite Hurricane Oscar Image

Hurricane Helene at peak

As shown in Figure 3.2, hurricane HELENE exhibits a more symmetric pressure field, with concentric pressure contours centred around the eye of the storm. This corresponds closely with the satellite image, which shows a well-organised cyclone with a clearly defined eye.

Use of this case with the Holland model is more appropriate, as the storm structure is relatively uniform.

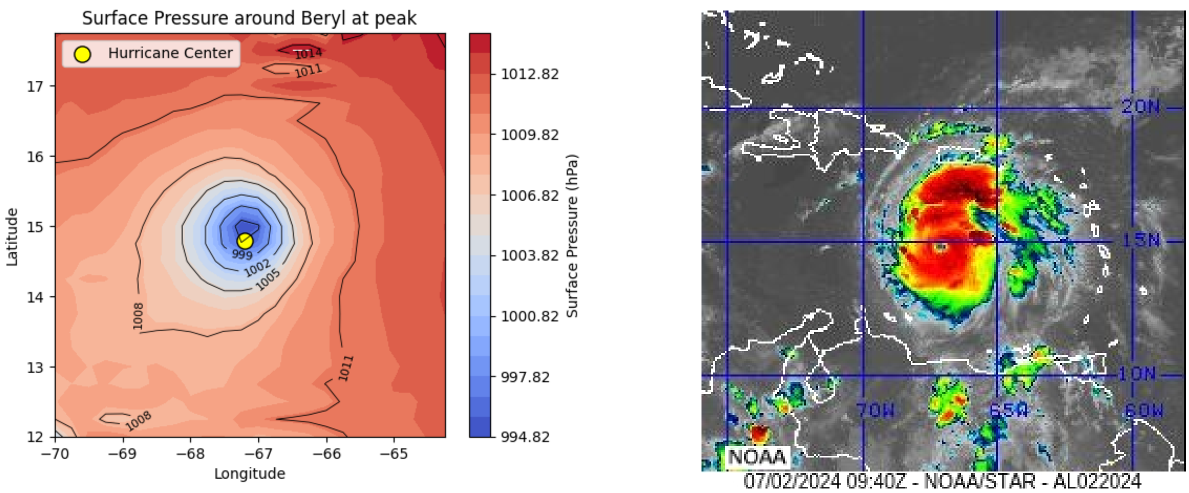


(a) Surface Pressure Field (b) GOES-16 Band 13 image

Figure 3.2: Comparison of Real Surface Pressure data and the Satellite Hurricane Helene Image

Hurricane Beryl at peak

Figure 3.3 illustrates a slightly asymmetric pressure field for Hurricane BERYL, with variations in the gradient around the storm centre. The satellite imagery shows a compact but somewhat irregular structure, indicating that environmental conditions may be affecting the hurricane's shape.



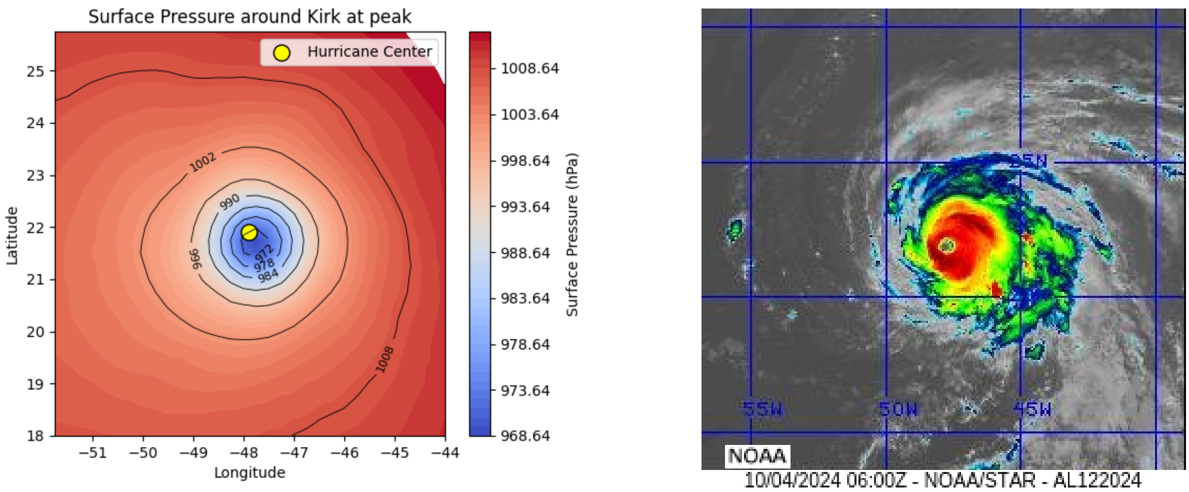
(a) Surface Pressure Field (b) GOES-16 Band 13 Image

Figure 3.3: Comparison of Real Surface Pressure data and the Satellite Hurricane Beryl Image

Hurricane Kirk at peak

In Figure 3.4, Hurricane KIRK shows noticeable asymmetry in both the pressure field and satellite imagery. The pressure gradients are uneven, and the cloud structure is clearly not circular.

This case highlights the limitations of symmetric parametric models such as H80 and H10, as they are unable to capture directional variability in the storm structure.



(a) Surface Pressure Field

(b) GOES-16 Band 13 Image

Figure 3.4: Comparison of Real Surface Pressure data and the Satellite Hurricane Kirk Image

3.1.3 Application of Pressure Fields to the Holland Model

In the standard Holland (2010) formulation, the environmental pressure P_n is assumed to be constant throughout the radial domain. However, analysis of ERA5 reanalysis data in the previous section shows that the environmental pressure is not spatially uniform, but instead varies with distance from the storm centre. This motivates a modification of the Holland model in which the environmental pressure is treated as a function of radius, $P_n = P_n(r)$.

Under this formulation, the pressure profile from becomes:

$$P_c = P_n - (P_n(r) - P_c)e^{-(r_{vm}/r)^b} \tag{3.1}$$

This modification represents a natural extension of the Holland model, allowing the addition of spatially varying atmospheric conditions derived from ERA5 data.

As a consequence, the parameter b is no longer determined only from a constant pressure difference, but is instead estimated by fitting the model to the observed radial pressure profile using nonlinear least squares.

Since the H80 model is formulated in terms of gradient level quantities, it is not directly compatible with the ERA5 surface pressure data. Hence, this stage of the analysis focuses exclusively on the H10 model.

To evaluate the impact of this modification, two versions of the H10 model were considered. The base model, which uses constant environmental pressure, and a modified model, which uses spatially varying pressure derived from ERA5.

For each hurricane, the radial pressure profile was constructed from ERA5 data (see Appendix A, pp. 46, 47, 50) and used to estimate the parameter b. The resulting wind profiles were then compared with those obtained under the constant pressure assumption.

In practice, the ERA5 pressure field, originally defined on a latitude–longitude grid, was first converted into radial distance from the storm centre using a spherical Earth approximation. The great circle distance was computed (see Appendix B, pp. 58) using the haversine formulation,

$$d = 2Rarcsin(\sqrt{sin^2(\frac{\Delta\phi}{2}) + cos(\phi_0)cos(\phi)sin^2(\frac{\Delta\lambda}{2})}) \tag{3.2}$$

where R is the Earth's radius, ϕ is latitude grid point from ERA5, ϕ_0 is latitude of the storm centre, $\Delta\phi$ is the difference between ϕ and ϕ_0, $\Delta\lambda$ is the difference between λ and λ_0, which are latitude grid point from ERA5 and latitude of the storm centre respectively [15]. This allowed the pressure field to be expressed in terms of radial distance (km) from the storm centre.

The data were then azimuthally averaged to obtain a mean radial pressure profile by binning values into 1 km intervals. Linear interpolation was applied within the interior of the dataset to ensure continuity between available data points, while no interpolation was performed at the inner and outer boundaries in order to avoid introducing assumptions where data coverage was incomplete (see Appendix B, pp. 58, 59). The resulting mean pressure profile was then used in a bounded nonlinear least squares fitting procedure to estimate the Holland b parameter (see Appendix A, pp. 47, 48).

3.1.4 H10 modelling with Pressure Fields

In the standard H10 implementation, the Holland b parameter is estimated from the maximum wind speed, central pressure, and assumed ambient pressure. In this modified version, that value is used only as an initial estimate, after which b is optimised by fitting the Holland pressure equation to observed radial pressure data. The resulting wind profile therefore, incorporates real pressure variation rather than relying on an assumed pressure, while still using the observed 34, 50, and 64 knot wind radii to constrain the radial decay of the wind field.

Hurricane Oscar

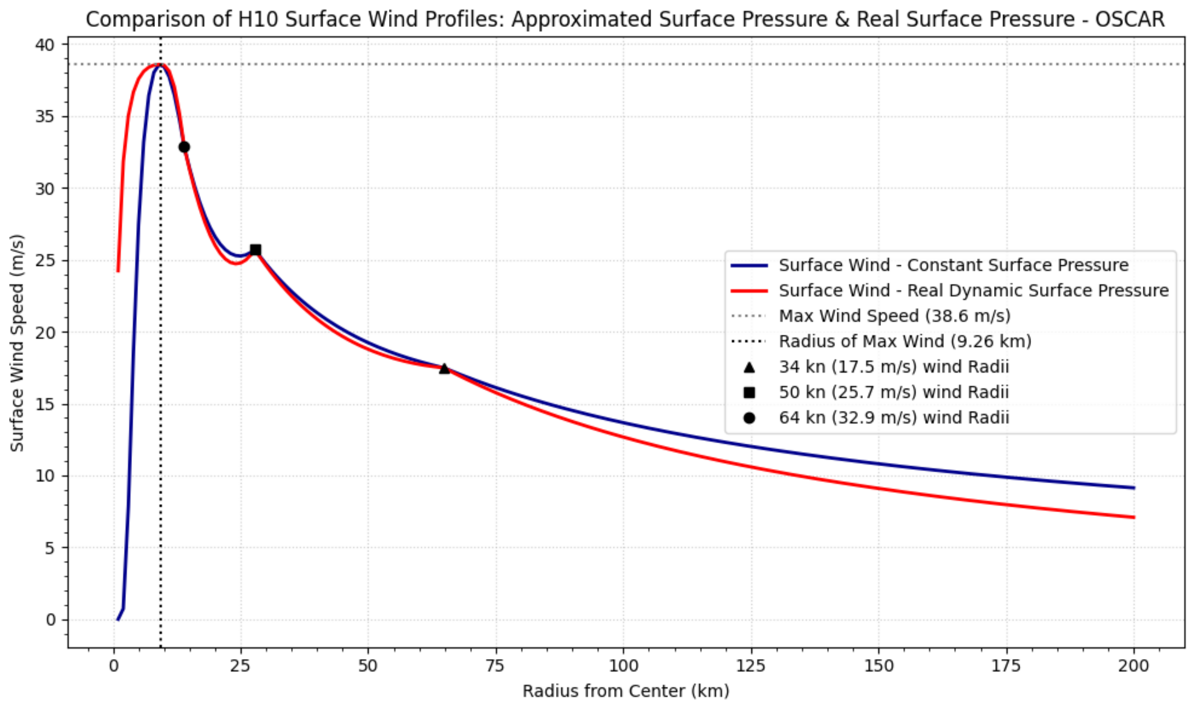


Figure 3.5: Wind Profile of Hurricane OSCAR at its peak intensity in 2024

Quantitative analysis Original H10 & Pressure Fitted H10 - OSCAR					
Interval (km)	H10 Mean (m/s)	H10 PF Mean (m/s)	RMSD (m/s)	Normalised RMSD (%)	Maximum Deviation (m/s)
0 - 5	6.8	31.9	25.57	132.02	31.08
5 - 55	25.66	25.7	1.66	6.46	10.11
55 - 105	15.74	15.21	0.61	3.97	1.07
105 - 200	10.80	9.1	1.71	17.27	2.06

In the Figure 3.5, we can observe differences occur within the eye and in the outer radial region. The pressure-fitted H10 model produces significantly higher wind speeds within the eye, indicating a steeper pressure gradient when real pressure variation is incorporated. In contrast, differences in the outer region are more moderate, with the pressure-fitted profile showing a slightly faster decay and lower wind speeds at larger radii compared to the constant-pressure case. Additionally, a lack of smoothness is evident in the transition between the 64 and 50 knot wind radii, where the profile shows a noticeable change in curvature, likely resulting from the interpolation between anchor points as well as averaging the pressure from the ERA5 grid. Overall, the incorporation of spatially varying pressure affects the eye structure with a noticeable 132% increase in wind speed of the Pressure-Fitted H10 model, while introducing minor but consistent differences in the outer wind field exhibiting up to 17% decrease compared to the original H10 model.

Hurricane Helene

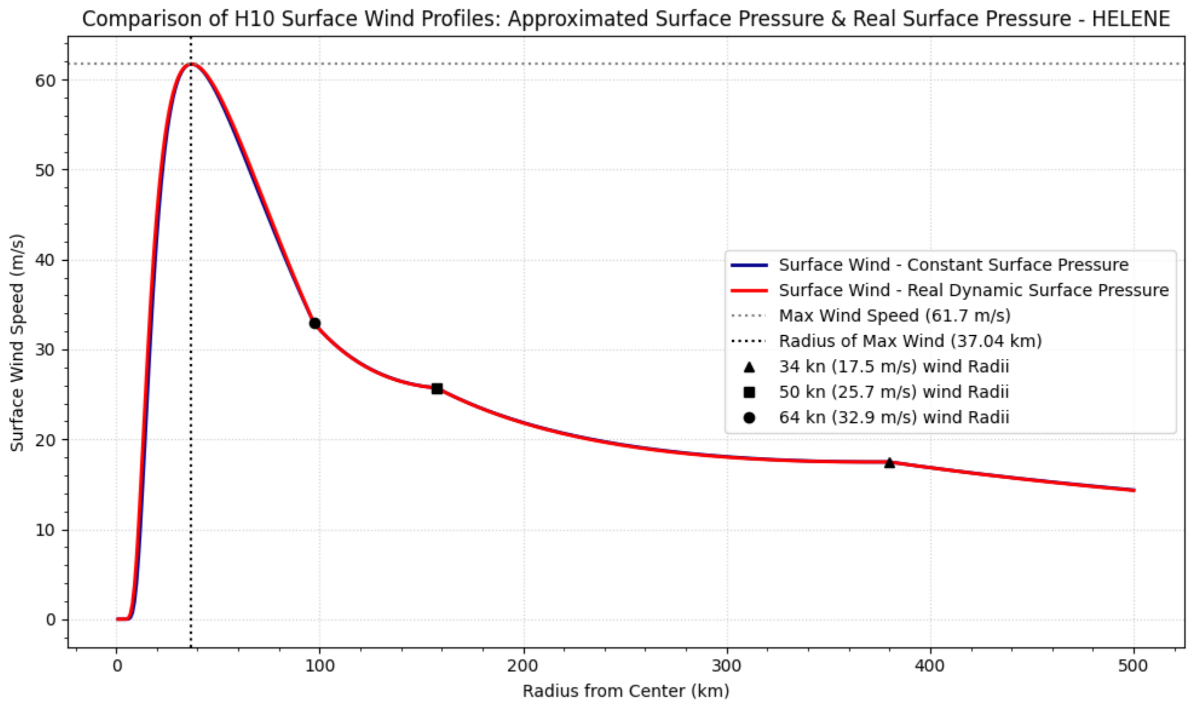


Figure 3.6: Wind Profile of Hurricane HELENE at its peak intensity in 2024

Quantitative analysis Original H10 & Pressure Fitted H10 - HELENE					
Interval (km)	H10 Mean (m/s)	H10 PF Mean (m/s)	RMSD (m/s)	Normalised RMSD (%)	Maximum Deviation (m/s)
0 - 20	11.35	13.42	2.67	21.58	4.20
20 - 70	56.04	56.53	0.72	1.28	2.49
70 - 120	35.82	35.99	0.27	0.76	0.54
120 - 220	24.33	24.31	0.03	0.13	0.05
220 - 320	18.9	18.84	0.05	0.29	0.05
320 - 500	16.39	16.36	0.02	0.17	0.04

For Hurricane Helene, in Figure 3.6, differences between the constant-pressure and pressure-fitted H10 models are almost nonexistent. However, a lack of smoothness is evident between the 64 and 50 knot wind radii.

Hurricane Beryl

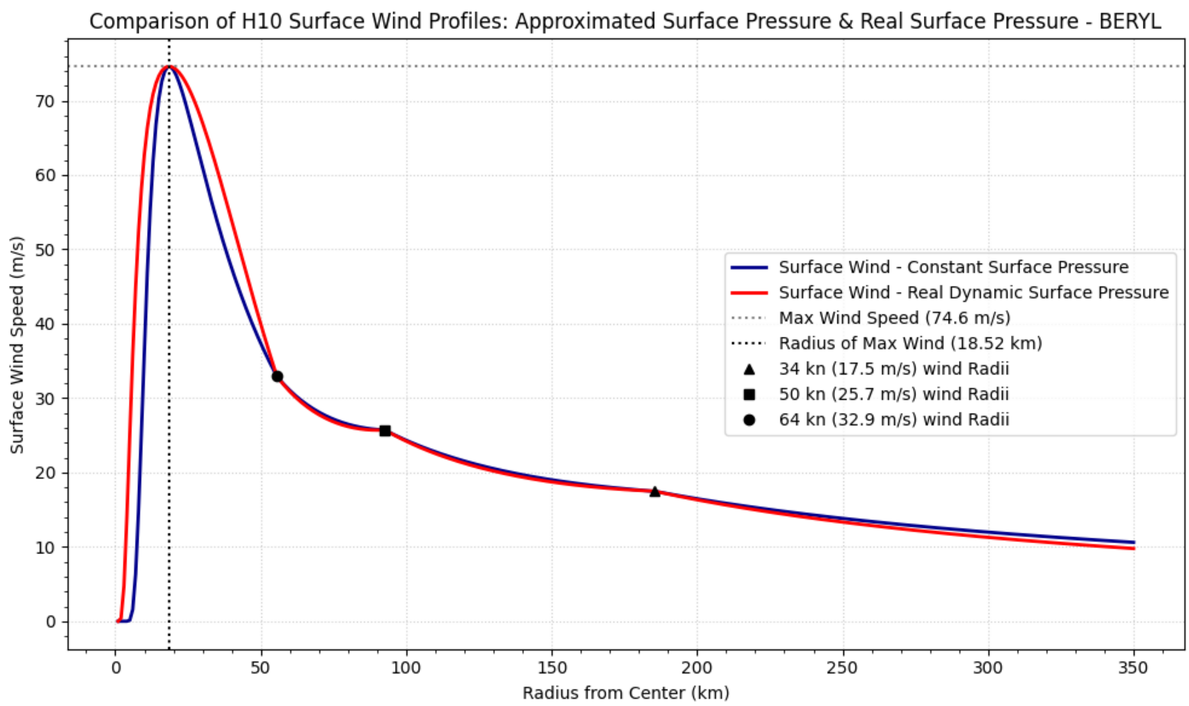


Figure 3.7: Wind Profile of Hurricane BERYL at its peak intensity in 2024

Quantitative analysis Original H10 & Pressure Fitted H10 - BERYL					
Interval (km)	H10 Mean (m/s)	H10 PF Mean (m/s)	RMSD (m/s)	Normalised RMSD (%)	Maximum Deviation (m/s)
0 - 10	5.39	26.16	25.73	163.02	38.52
10 - 60	52.68	57.34	6.63	12.06	26.01
60 - 110	26.37	26.15	0.23	0.9	0.35
110 - 210	18.8	18.6	0.23	1.23	0.32
210 - 350	13.54	13.02	0.53	4.05	0.72

For Hurricane Beryl, the largest differences between the constant-pressure and pressure-fitted H10 models occur within the eye, where the pressure-fitted profile produces significantly higher with 163% increase in wind speeds. Beyond the inner region, the profiles converge, with only minor deviations observed across the outer radii, suggesting that the inclusion of spatially varying pressure has a limited influence on the broader wind field structure. Overall, the modification primarily affects the inner-core structure, while the outer wind field remains largely unchanged.

Hurricane Kirk

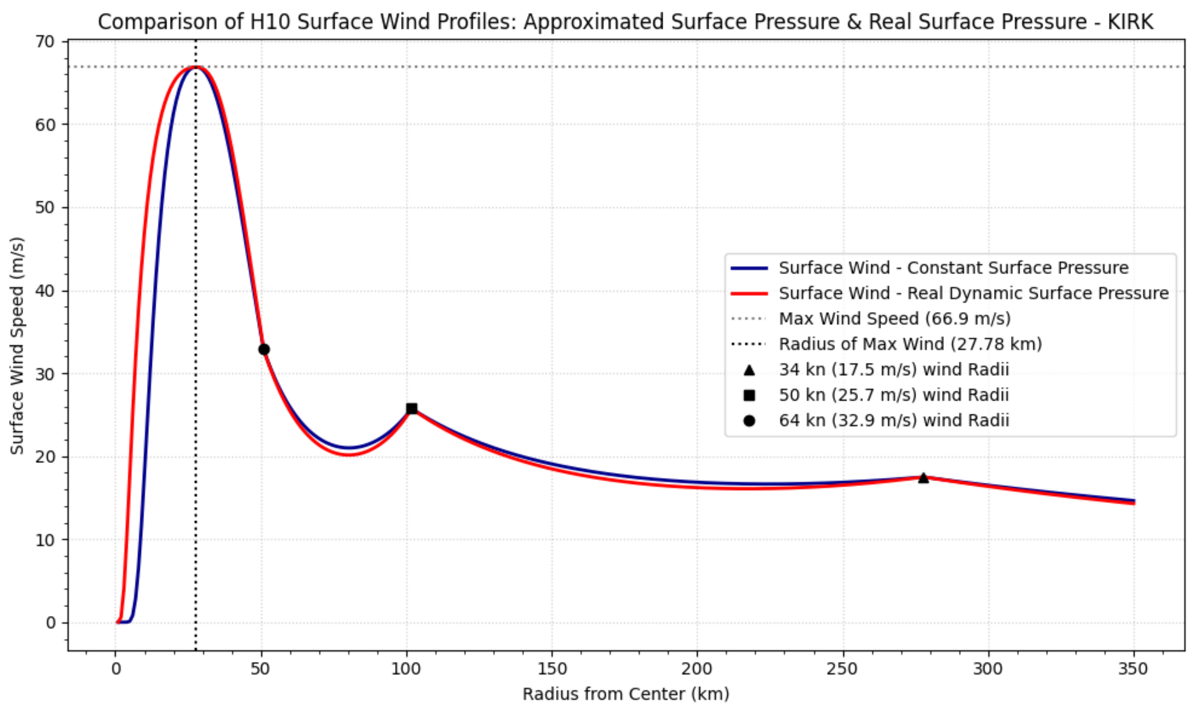


Figure 3.8: Wind Profile of Hurricane KIRK at its peak intensity in 2024

Quantitative analysis Original H10 & Pressure Fitted H10 - KIRK					
Interval (km)	H10 Mean (m/s)	H10 PF Mean (m/s)	RMSD (m/s)	Normalised RMSD (%)	Maximum Deviation (m/s)
0 - 15	12.28	15.47	4.08	29.46	6.34
15 - 65	48.73	49.12	0.86	1.76	3.65
65 - 115	22.8	22.69	0.12	0.56	0.17
115 - 215	18.69	18.58	0.11	0.61	0.13
215 - 350	16.80	16.75	0.06	0.36	0.11

Looking at Figure 3.8 representing Hurricane Kirk, the most noticeable differences between the constant-pressure and pressure-fitted H10 models occur within the eye, where the pressure-fitted profile produces moderately higher wind speeds, reflecting a steeper local pressure gradient. Outside this region, the two profiles remain closely aligned, with only minimal deviations across the outer radii. A lack of smoothness is again evident between the 64 and 50 knot wind radii.

Pressure Fitting Results

Across all cases, with the exception of Hurricane Helene, the inclusion of spatially varying pressure fields leads to notable changes within the eye of the storm, indicating a stronger sensitivity of the eye structure to pressure variation. However, the overall shape of the wind field remains largely unchanged, including the persistence of localised dips between the 64 and 50 knot wind radii. This suggests that these features are primarily driven by the imposed radial constraints rather than the pressure formulation itself. Consequently, a logical next step would be to incorporate the full directional information available in the HURDAT2 dataset, utilising based on quadrant wind radii 34, 50, and 64 knots, and combine these with the spatially varying pressure field derived from ERA5 to better capture asymmetries in the wind structure.

3.1.5 Extension to Asymmetric Wind Profiles

The previous sections demonstrated that incorporating spatially varying pressure fields into the H10 model results in only minor changes to the overall wind profile. However, analysis of ERA5 pressure fields and satellite imagery revealed that real hurricanes exhibit significant asymmetry, which is not captured by the standard symmetric formulation of the Holland model.

To address this limitation, the H10 model was extended to incorporate directional variability by constructing separate wind profiles for each quadrant: North-East, North-West, South-East, South-West.

For each quadrant, a separate radial pressure profile $P_n^{(q)}(r)$ was derived from ERA5 data, where $q \in \{\text{NE, NW, SE, SW}\}$ (see Appendix B, pp. 60, 61). Similarly, quadrant-specific wind radii (34, 50, and 64 knots) were used to define the anchor points for the parameter $x(r)$.

The modified H10 model was then applied independently to each quadrant:

$$V^{(q)}(r) = V_m \left[\left(\frac{r_{v_m}}{r} \right)^{b^{(q)}} e^{\left(1 - \left(\frac{r_{v_m}}{r} \right)^{b^{(q)}} \right)} \right]^{x^{(q)}(r)} \tag{3.3}$$

where the parameters $b^{(q)}$ and $x^{(q)}(r)$ are estimated separately for each quadrant using the corresponding pressure and wind data.

The resulting quadrant-based wind profiles are shown in the figures presented below, where each quadrant is plotted alongside the symmetric constant pressure H10 model.

Hurricane Oscar

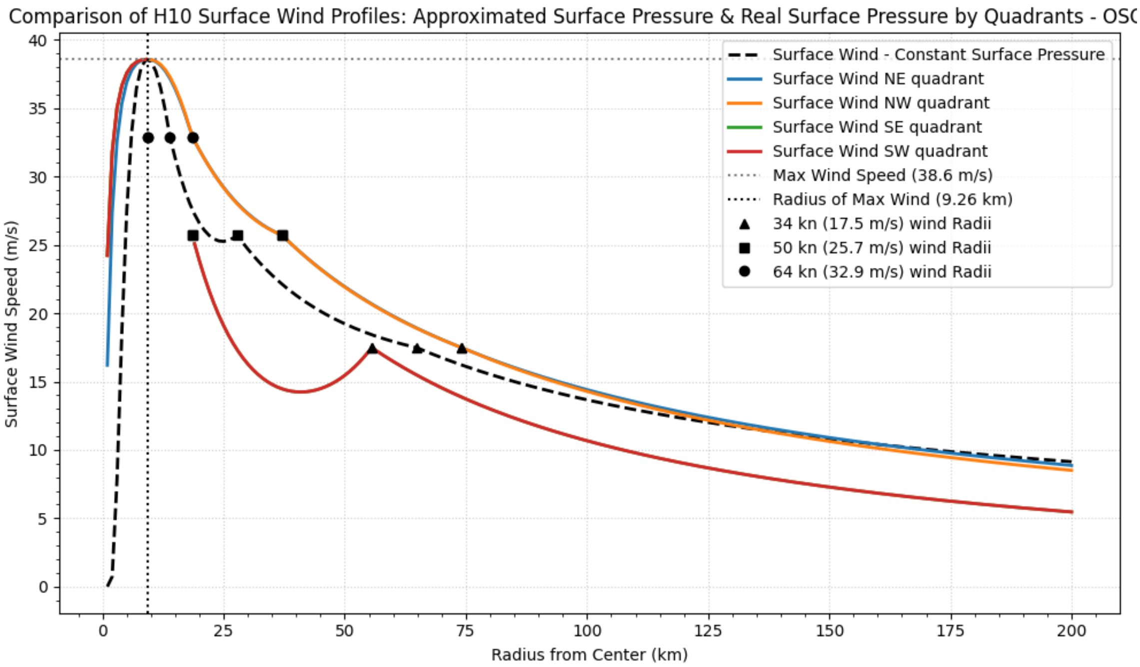


Figure 3.9: Wind profile of Hurricane Oscar at peak intensity

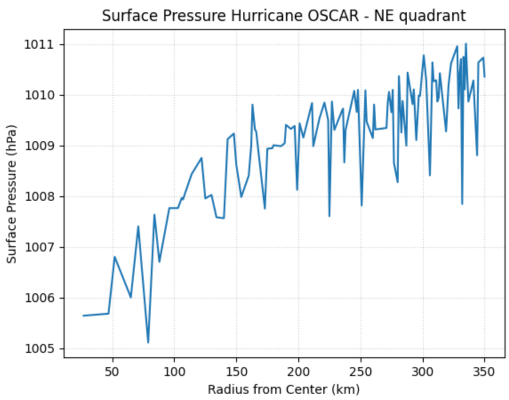
For Hurricane Oscar, notable inconsistencies between the wind profiles based on quadrants (Figure 3.9) can be directly linked to the irregular pressure fields observed in each quadrant (Figure 3.10). The pressure profiles derived from ERA5 present significant noise, the SE and SW quadrants in particular, where there are evident sharp fluctuations and non-monotonic behaviour. This lack of smoothness translates into the wind profiles, contributing to local dips and deviations, specifically between the 64 and 50 knot radii.

In the southern quadrants, an additional limitation arises where the 64 knot wind radius is located at the same distance as the eyewall, by the data from HURDAT2, effectively coinciding with the radius of maximum wind. This disrupts the model formulation, resulting in a loss of continuity in the wind profile.

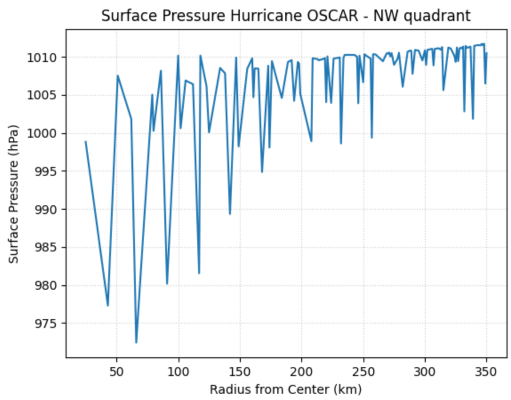
The cause of this chaotic pressure structure can be related to the geographical positioning of the storm at peak intensity, as shown in the satellite image (Figure 3.4(b)) and the corresponding pressure field (Figure 3.1), where the system is located between two islands. This configuration introduces strong spatial variability in the pressure field, with distinct gradients and anomalies across relatively short distances.

And lastly, despite dividing the domain into quadrants, the pressure is still azimuthally averaged within each quadrant, representing a 45 sector that includes diverse conditions, as illustrated in Figure 3.1. This averaging smooths but does not eliminate the underlying diversity, leading to inconsistencies in the derived radial profiles.

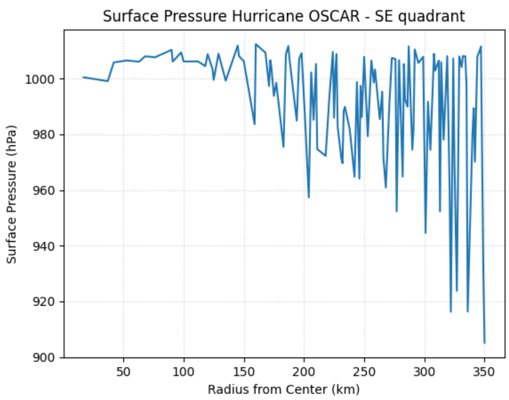
Consequently, the assumption of radial symmetry breaks down, resulting in inconsistencies in the fitted wind profiles. Overall, these findings indicate that the Holland model, in its standard symmetric formulation, is not well suited for representing storms with strongly asymmetric and environmentally disrupted pressure fields such as Hurricane Oscar.



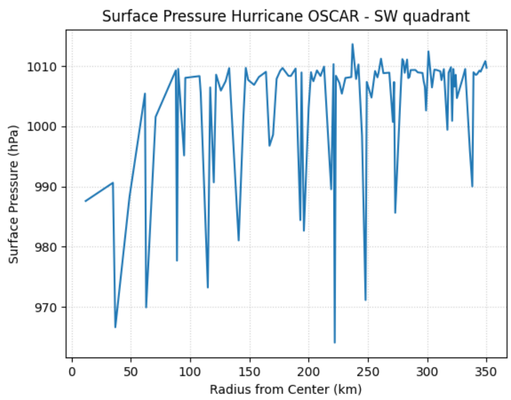
(a) North-East quadrant



(b) North-West quadrant



(c) South-East quadrant



(d) South-West quadrant

Figure 3.10: Wind profile and quadrant-based ERA5 surface pressure profiles for Hurricane Oscar at peak intensity.

Hurricane Helene

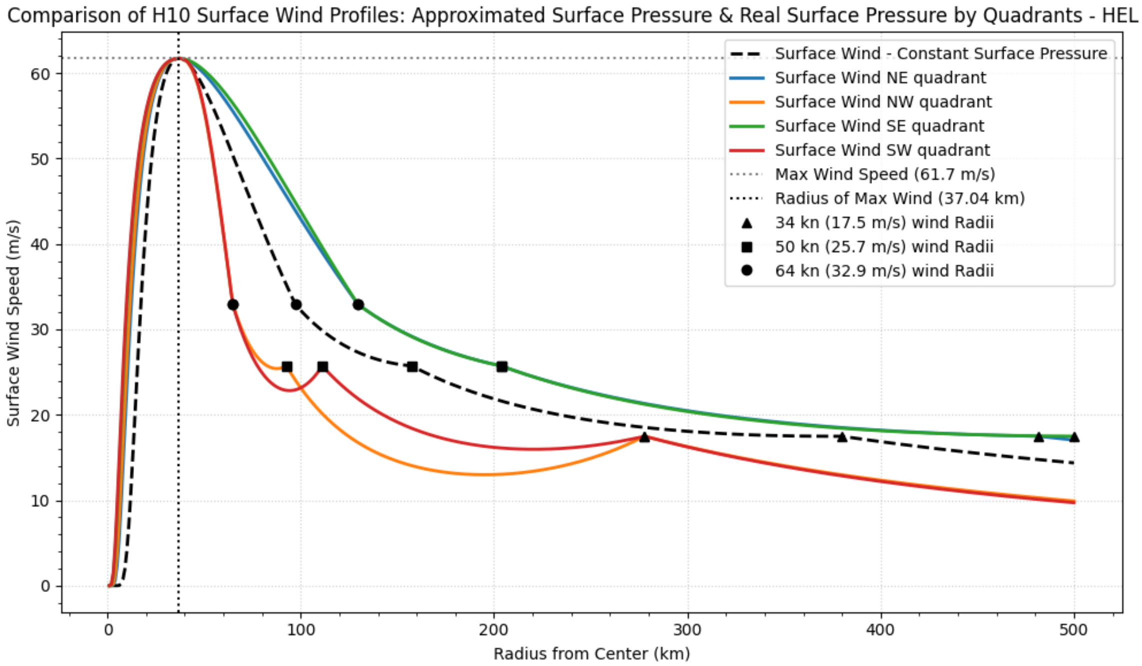


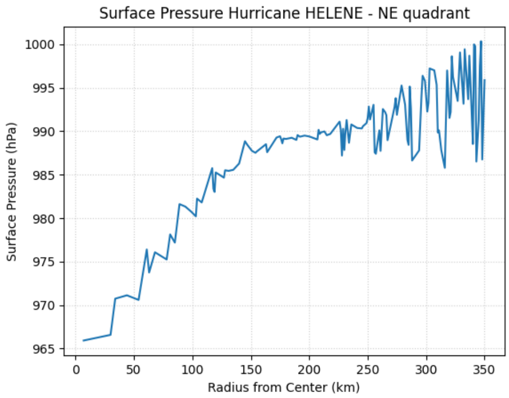
Figure 3.11: Wind profile of Hurricane Oscar at peak intensity

For Hurricane Helene, the wind profiles based on quadrant (Figure 3.11) show a clear dependence on the underlying pressure structure (Figure 3.12).
In North-East and South-East quadrants the wind profiles present a close to linear transition between the radius of maximum wind and the 64 knot radii, reflecting the relatively noisy pressure fields in these regions. As shown in the pressure map and satellite image (Figure 3.2), this behaviour is associated with the storm's proximity to landfall, where interaction with the coastline introduces irregular gradients in the pressure field due to pressure decrease related to surface elevation which was discussed earlier above.

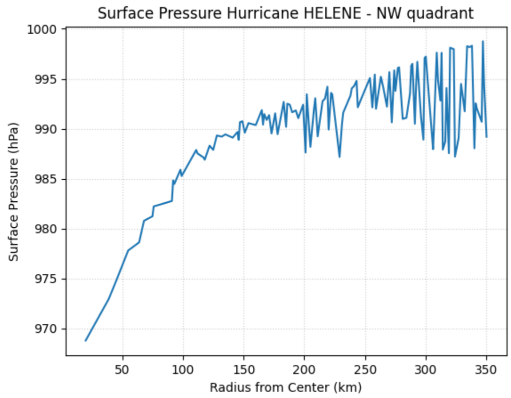
In contrast, the North-West and South-West quadrants are significantly more irregular, with pronounced dips in wind speed within the first two radial intervals beyond the eyewall. These features correspond closely to the highly variable and noisy pressure profiles observed in these quadrants, particularly near the storm centre for the South-West quadrant. For Noth-West as shown in the pressure field and satellite imagery (Figure 3.2), this behaviour is likely influenced by land interaction during imminent landfall, which disrupts the pressure curve and introduces asymmetry.

Overall, the modified approach yields improved performance in two out of four quadrants, South-

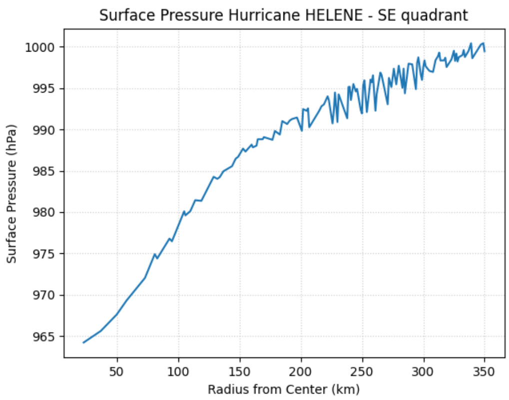
East and North-East, where smoother wind profiles are obtained, and the model aligns more closely with the observed wind radii. For the remaining two quadrants, such as South-West and North-West, the discrepancies suggest that further improvement could be achieved by obtaining more azimuthally resolved wind data, reducing the need to average pressure over large angular sectors.



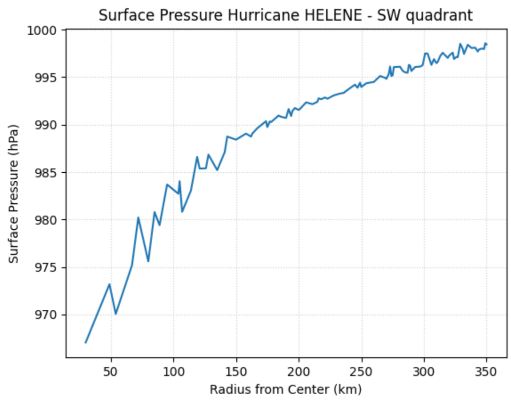
(a) North-East quadrant



(b) North-West quadrant



(c) South-East quadrant



(d) South-West quadrant

Figure 3.12: Wind profile and quadrant-based ERA5 surface pressure profiles for Hurricane Helene at peak intensity.

Hurricane Beryl

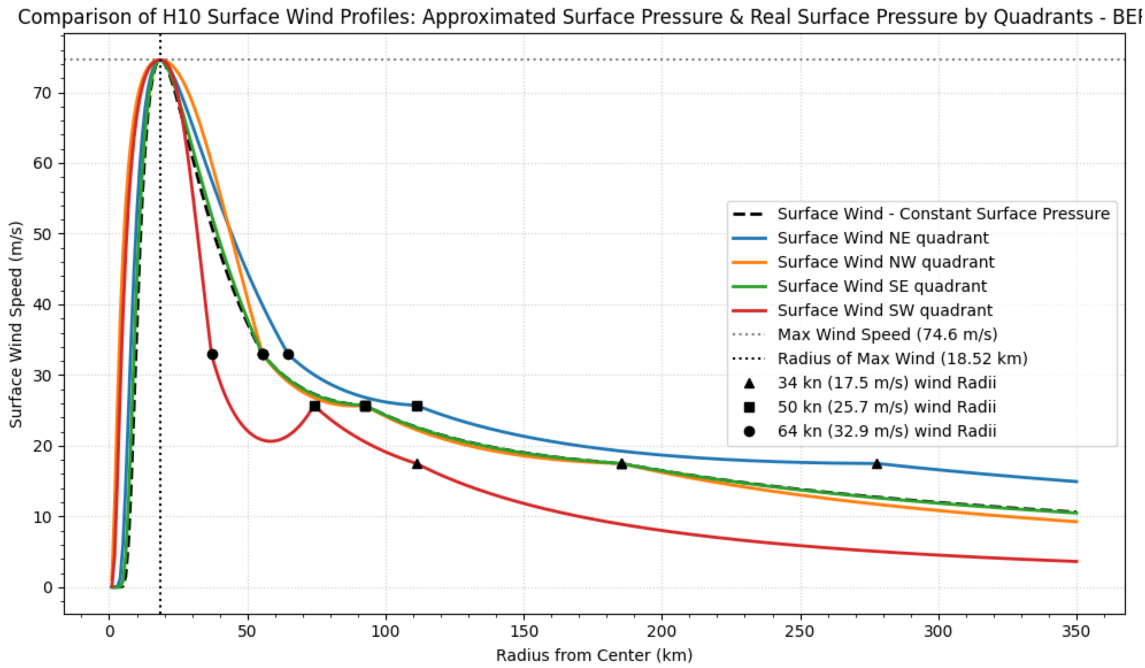
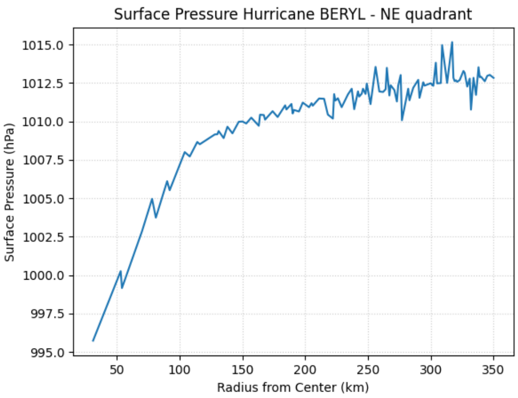


Figure 3.13: Wind Profile of Hurricane Beryl at its peak intensity

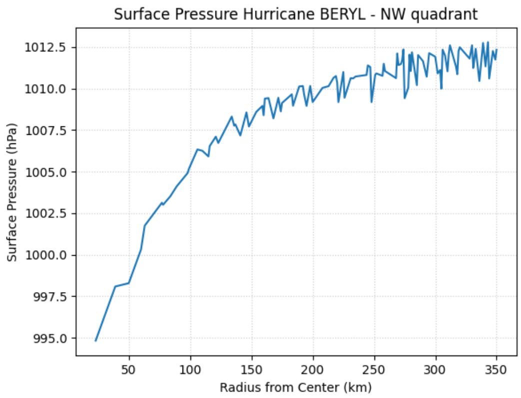
For Hurricane Beryl, the wind profiles based on quadrants (Figure 3.13) show generally consistent behaviour across most sectors, with the North-East, North-West, and South-East quadrants presenting relatively smooth profiles. These quadrants display slight curvature between the wind radii and align closely with the observed wind speed points, reflecting the comparatively smooth pressure distributions seen in the corresponding ERA5 profiles (Figure 3.14).

In contrast, the South-West quadrant presents a pronounced local dip between the 64 and 50 knot radii, followed by a smooth recovery at larger radii. While this irregularity is not strongly evident in the South-West radial pressure profile alone (Figure 3.14(d)), the pressure field (Figure 3.3(a)) reveals that this behaviour is characterised by relatively lower pressure compared to other sectors. This asymmetry in the pressure field likely highlights the local pressure gradient, acting as a catalyst for the observed deviation in the wind profile.

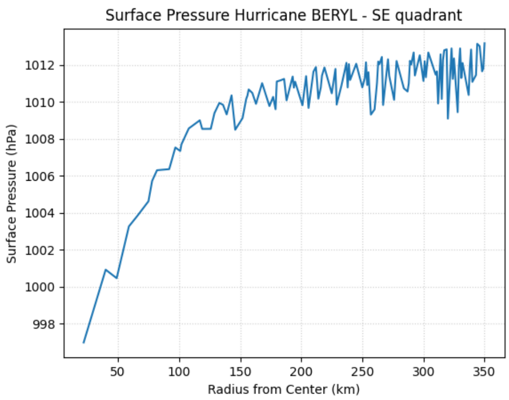
Overall, the results demonstrate that while the modified model performs well in regions with smooth pressure variation, local anomalies in the pressure field can still introduce significant deviations in the derived wind structure.



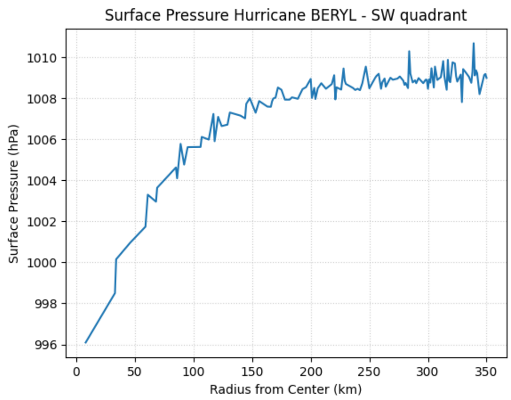
(a) North-East quadrant



(b) North-West quadrant



(c) South-East quadrant



(d) South-West quadrant

Figure 3.14: Quadrant-based ERA5 surface pressure profiles for Hurricane Beryl at peak intensity.

Hurricane Kirk

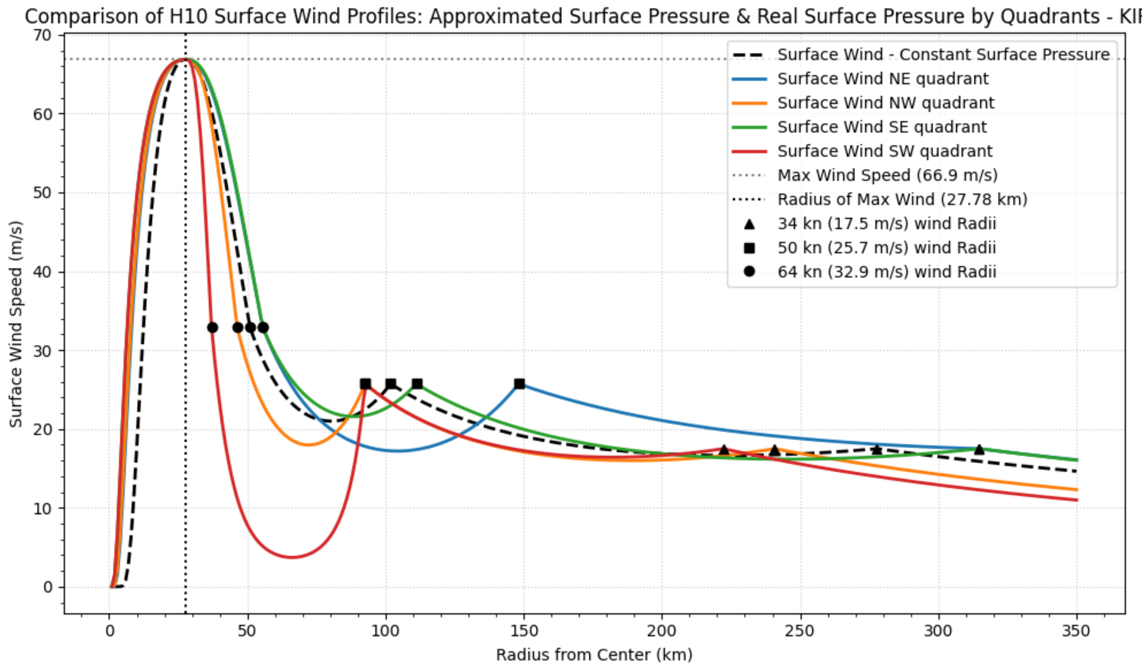


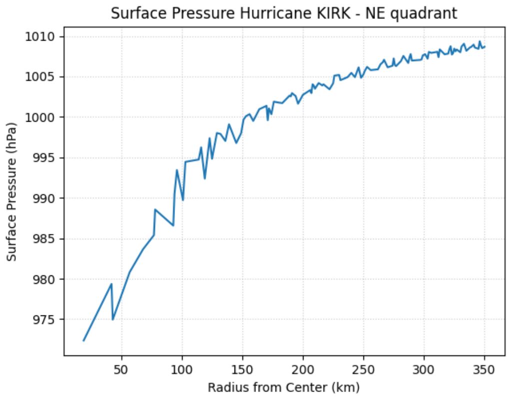
Figure 3.15: Wind Profile of Hurricane Kirk at its peak intensity

Hurricane Kirk presents a notably more complex and less consistent case compared to the previous storms, with obvious irregularities across multiple quadrants (Figure 3.15).

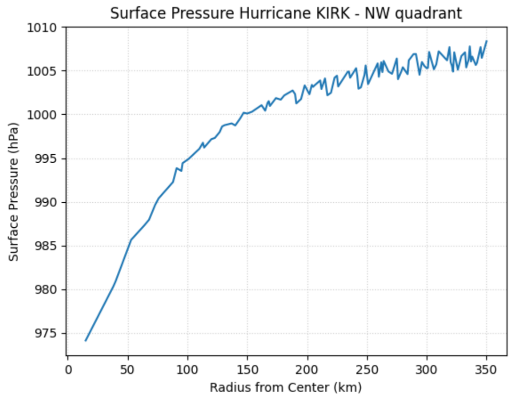
The most significant anomaly occurs in the SW quadrant, where a sharp local dip in the wind profile is observed between the 64 and 50 knot radii. This behaviour can be directly linked to the highly noisy first half of the pressure profile in this quadrant (Figure 3.16(d)), where substantial variability in the first 100 km disrupts the pressure gradient and translates into the wind structure.

Similar, though less severe, deviations are present in the North-East and South-East quadrants, where local fluctuations in the pressure data correspond to smaller dips in the wind profiles. In contrast, the North-West quadrant presents a relatively smooth pressure profile, with no clear anomalies visible in the pressure field (Figure 3.4(a)). However, despite this, the resulting wind profile still shows noticeable deviations and lacks smoothness between the real wind speed observations, suggesting that factors beyond the radial pressure distribution, such as limited azimuthal resolution of wind radii or linear interpolation constraints, may be influencing the model output.

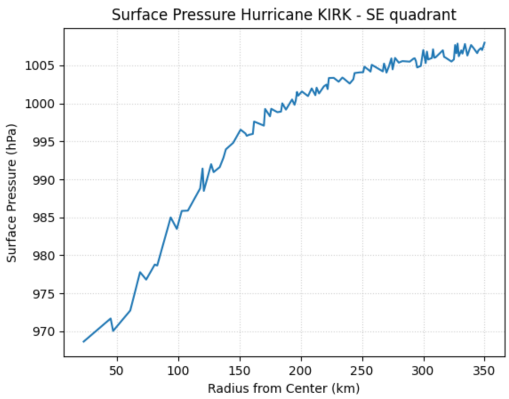
Overall, this case highlights the limitations of the current approach and indicates that additional observational data, particularly with improved directional resolution, are required to fully capture and interpret the wind structure in such complex scenarios with Holland model.



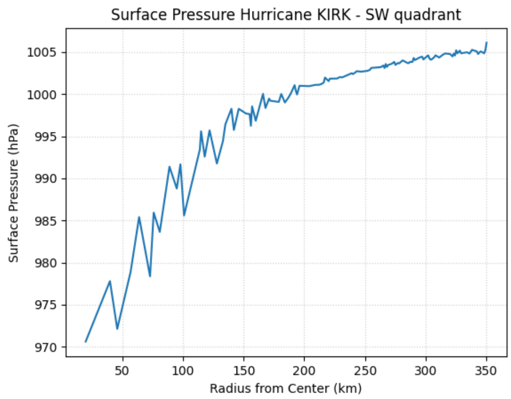
(a) NE quadrant



(b) NW quadrant



(c) SE quadrant



(d) SW quadrant

Figure 3.16: Quadrant-based ERA5 surface pressure profiles for Hurricane Kirk at peak intensity.

3.2 Conclusion

This study has demonstrated that the Holland parametric model, even when applied with limited observational data, is capable of producing smooth wind profiles that broadly align with recorded wind radii. In cases where the pressure field is relatively consistent, with no land in proximity and free from strong anomalies, the model provides a reasonable representation of the wind structure, supporting its continued use in simplified modelling applications.

The introduction of spatially varying pressure fields enabled the extension of the model toward asymmetric wind profile representation. However, the results show that the effectiveness of this approach is strongly dependent on the quality and resolution of the input data. In several cases, particularly where the pressure field exhibited significant noise or spatial deviations, the resulting wind profiles contained discontinuities and unrealistic features. This highlights the sensitivity of the model to input data and the limitations of applying radial averaging over large azimuthal sectors.

While the results indicate that constructing asymmetric wind profiles using the Holland framework is feasible, achieving reliable and physically consistent outputs requires more detailed observational data, particularly with improved azimuthal resolution of wind radii and pressure fields. Furthermore, alternative methods for processing and averaging pressure field data may offer improvements over the current approach and represent a promising direction for future work.

Overall, the findings suggest that while the Holland model remains a useful and computationally efficient tool, its application to complex, asymmetric storm environments is limited without sufficient high-quality data and more advanced treatment of spatial variability.

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Appendix A

Jupyter Notebook Wind Profile Modelling

Python Packages in use

In [4]:
```
import pandas as pd
import re
import numpy as np
import matplotlib.pyplot as plt
import importlib
import os
import math
from scipy.optimize import curve_fit
import statistics as stat
```

Processing the HURDAT2 dataset

In [15]:
```
file_path = "data/hurdat2-1851-2024-040425.csv"

records = []
storm_id, storm_name = None, None

with open(file_path, "r") as f:
    for line in f:
        parts = [p.strip() for p in line.split(",")]
        if len(parts) == 4:
            storm_id, storm_name, count, nan = parts
            continue

        if len(parts) > 4:
            date, time, record_id, status, lat, lon, maxwind, minpressure, thirtyne
            records.append([
                storm_id, storm_name, date, time, record_id, status,
                lat, lon, maxwind, minpressure, thirtyne, thirtyse, thirtysw,
                thirtynw, fivene, fivese, fivesw, fivenw, sixne, sixse, sixsw,
                sixnw, radmaxwind
            ])

# Define column names
cols = ["SID", "Name", "Date", "Time", "RecordID", "Status", "Latitude", "Longitude
        "MaxWind(kn)", "MinPressure", "34NE", "34SE", "34SW", "34NW", "50NE",
        "50SE", "50SW", "50NW", "64NE", "64SE", "64SW", "64NW", "RadiusOfMaxWind"]

df = pd.DataFrame(records, columns=cols)

df = df.replace(["", " ", "-999"], pd.NA)

quad_cols = [
    '34NE','34SE','34SW','34NW',
    '50NE','50SE','50SW','50NW',
    '64NE','64SE','64SW','64NW'
]
df[quad_cols] = df[quad_cols].apply(pd.to_numeric, errors = 'coerce')
```

```
df["Date"] = pd.to_datetime(df["Date"], format="%Y%m%d").dt.date
df["Time"] = pd.to_datetime(df["Time"].str.zfill(4), format="%H%M").dt.time
```

In [19]:
```
beryl_df = df[(df["Name"]=="BERYL") & (df["Status"]=="HU") & (pd.notna(df["RadiusOf
```

Final obtained data on the Hurricane in peak intensity

In [23]:
```
row = beryl_df.iloc[11]
row
```

Out[23]:
```
index                      54777
SID                     AL022024
Name                       BERYL
Date                  2024-07-02
Time                    09:45:00
RecordID                       I
Status                        HU
Latitude                   14.8N
Longitude                  67.2W
MaxWind(kn)                  145
MinPressure                  932
34NE                       150.0
34SE                       100.0
34SW                        60.0
34NW                       100.0
50NE                        60.0
50SE                        50.0
50SW                        40.0
50NW                        50.0
64NE                        35.0
64SE                        30.0
64SW                        20.0
64NW                        30.0
RadiusOfMaxWind               10
Name: 11, dtype: object
```

Pressure data from ERA5 notebook

In [26]:
```
PN = pd.read_csv("data/beryl_pressure_average.csv")
PN = PN['SP']*100
```

In [30]:
```
r_data = np.arange(1, len(PN)+1) * 1000
p_data = PN.values
r_data, p_data = r_data[~np.isnan(p_data)], p_data[~np.isnan(p_data)]
```

Unit conversions

In [33]:
```python
# Unit Conversions
KNOTS_TO_M_S = 0.514444
NMI_TO_M = 1852.0
NMI_TO_KM = 1.852
HPA_TO_PA = 100.0
BL_REDUCTION_FACTOR = 0.8

rad34_v = stat.median([float(row['34NE']), float(row['34SE']), float(row['34NW']),
rad34_v_m = rad34_v * NMI_TO_M
rad34_v_ms = 34.0 * KNOTS_TO_M_S

rad50_v = stat.median([float(row['50NE']), float(row['50SE']), float(row['50NW']),
rad50_v_m = rad50_v * NMI_TO_M
rad50_v_ms = 50.0 * KNOTS_TO_M_S

rad64_v = stat.median([float(row['64NE']), float(row['64SE']), float(row['64NW']),
rad64_v_m = rad64_v * NMI_TO_M
rad64_v_ms = 64.0 * KNOTS_TO_M_S
```

H80 wind profile model

In [41]:
```
def holland80(row, r_max=500):
    """
    ym  : maximum wind (m/s)
    rym : radius of max wind (km)
    pcs : central pressure (hPa)
    pns : ambient pressure (hPa)
    """
    ym = float(row['MaxWind(kn)']) * KNOTS_TO_M_S  # Vmax: knots to m/s
    ym = ym * 1.25 # H80 model uses gradient wind speed instead of surface speed
    pcs = float(row['MinPressure']) * HPA_TO_PA       # Pc: hPa to Pa
    rym = float(row['RadiusOfMaxWind']) * NMI_TO_M  # Rmax: nmi to m
    pns = 101335
    # Approximate air density
    rho = 1.15
    # Pressure drop
    dp = pns - pcs
    # b parameter
    b = (ym**2 * rho * np.e) / (dp)
    print(b)
    # Radius array - 1 km spacing
    r_km = np.arange(1, r_max + 1)
    r_m  = r_km * 1000
    # Wind profile
    for i in range(len(r_m)):
        if r_m[i] < rym:
            x = 1
        else:
            x = 0.5
        term = (rym / r_m)**b
        wind = ym * (term * np.exp(1 - term))**x
    wind = wind/1.25
    return r_km.tolist(), wind.tolist()
```

In [45]:
```
H80_r_km, H80_wind_ms = holland80(row)
h80_df = pd.DataFrame({
    'Distance_km': H80_r_km,
    'Wind_ms': H80_wind_ms
})
```

```
3.340927168591843
```

H10 wind profile model

In [48]:
```
def x_anchor(ym, rym, rad_v_m, rad_v_ms, b):
    ratio = rym / rad_v_m
    denominator = (b * np.log(ratio)) + 1.0 - (ratio)**b
    return np.log(rad_v_ms / ym) / denominator
```

In [50]:
```
def interp_x(r, r1, x1, r2, x2):
    return x1 + (x2 - x1) * ((r - r1) / (r2 - r1))
```

In [52]:
```
def holland10(row, r_max = 500):
    """
```

```
    Simple Holland (2010) wind profile for one input.

    ym  : maximum wind (m/s)
    rym : radius of max wind (km)
    pcs : central pressure (hPa)
    pns : ambient pressure (hPa)
    """
    ym = float(row['MaxWind(kn)']) * KNOTS_TO_M_S  # Vmax: knots to m/s
    pcs = float(row['MinPressure']) * HPA_TO_PA       # Pc: hPa to Pa
    rym = float(row['RadiusOfMaxWind']) * NMI_TO_M  # Rmax: nmi to m
    pns = 101335

    # Approximate air density with use of surface pressure approximation
    rho = 1.15

    # Pressure drop
    dp = pns - pcs

    # Holland b parameter
    b = (ym**2 * rho * np.e) / (dp)
    print(b)

    # Radius array (1 km spacing)
    r_km = np.arange(1, r_max + 1)
    r_m  = r_km * 1000
    wind = np.zeros_like(r_m, dtype=float)

    x64 = x_anchor(ym, rym, rad64_v_m, rad64_v_ms, b)
    x50 = x_anchor(ym, rym, rad50_v_m, rad50_v_ms, b)
    x34 = x_anchor(ym, rym, rad34_v_m, rad34_v_ms, b)

    # Wind profile
    i = 0
    for i in range(len(r_m)):
        if r_m[i] <= rym:
            x = 0.5
        elif r_m[i] <= rad64_v_m:
            x = interp_x(r_m[i], rym, 0.5, rad64_v_m, x64)
        elif r_m[i] <= rad50_v_m:
            x = interp_x(r_m[i], rad64_v_m, x64, rad50_v_m, x50)
        elif r_m[i] <= rad34_v_m:
            x = interp_x(r_m[i], rad50_v_m, x50, rad34_v_m, x34)
        else:
            x = x34
        term = (rym / r_m[i])**b
        wind[i] = ym * (term * np.exp(1 - term))**x

    return r_km.tolist(), wind.tolist()
```

Enchanced H10 wind profile model with Pressure Fitting

In [55]:
```python
def holland10_dynamic_pressure(r_data, p_data, row, r_max = 500, quad = None):
    """
    Simple H10 wind profile for one input.

    ym  : maximum wind (m/s)
    rym : radius of max wind (km)
    pcs : central pressure (hPa)
    pns : ambient pressure (hPa)
    """
    ym = float(row['MaxWind(kn)']) * KNOTS_TO_M_S  # Vmax: knots to m/s
    pcs = float(row['MinPressure']) * HPA_TO_PA       # Pc: hPa to Pa
    rym = float(row['RadiusOfMaxWind']) * NMI_TO_M  # Rmax: nmi to m
    pns = 101335

    if quad != None:
        rad64_v_m = row[f'64{quad}']*NMI_TO_M
        rad50_v_m = row[f'50{quad}']*NMI_TO_M
        rad34_v_m = row[f'34{quad}']*NMI_TO_M
    else:
        rad64_v_m = rad64_v * NMI_TO_M
        rad50_v_m = rad50_v * NMI_TO_M
        rad34_v_m = rad34_v * NMI_TO_M

    # Approximate air density at the gradient level constant 1.15
    rho = 1.15

    # Pressure drop
    dp = pns - pcs

    # Radius array (1 km spacing)
    r_km = np.arange(1, r_max + 1)
    r_m  = r_km * 1000
    wind = np.zeros_like(r_m, dtype=float)

    # Holland b parameter
    b_initial = (ym**2 * rho * np.e) / (dp)
    print(b_initial)

    def holland_pressure_b(r, b):
        p = pcs + (pns - pcs) * np.exp(-(rym / r)**b)
        return p

    b_fit, _ = curve_fit(
        holland_pressure_b,
        r_data,
        p_data,
        p0=[b_initial],
        bounds=(0.5,3.0)
    )

    b = b_fit[0]
    print(b)

    x64 = x_anchor(ym, rym, rad64_v_m, rad64_v_ms, b)
    x50 = x_anchor(ym, rym, rad50_v_m, rad50_v_ms, b)
```

```
    x34 = x_anchor(ym, rym, rad34_v_m, rad34_v_ms, b)

    # Wind profile
    i = 0
    for i in range(len(r_m)):
        if r_m[i] <= rym:
            x = 0.5
        elif r_m[i] <= rad64_v_m:
            x = interp_x(r_m[i], rym, 0.5, rad64_v_m, x64)
        elif r_m[i] <= rad50_v_m:
            x = interp_x(r_m[i], rad64_v_m, x64, rad50_v_m, x50)
        elif r_m[i] <= rad34_v_m:
            x = interp_x(r_m[i], rad50_v_m, x50, rad34_v_m, x34)
        else:
            x = x34
        term = (rym / r_m[i])**b
        wind[i] = ym * (term * np.exp(1 - term))**x

    return r_km.tolist(), wind.tolist()
```

In [57]:
```
H10_r_km, H10_wind_ms = holland10(row)
h10_df = pd.DataFrame({
    'Distance_km': H10_r_km,
    'Wind_ms': H10_wind_ms
})
```

```
2.13819338789878
```

In [59]:
```
H10_r_km_DP, H10_wind_ms_DP = holland10_dynamic_pressure(r_data, p_data, row)
h10_DP_df = pd.DataFrame({
    'Distance_km': H10_r_km_DP,
    'Wind_ms': H10_wind_ms_DP
})
```

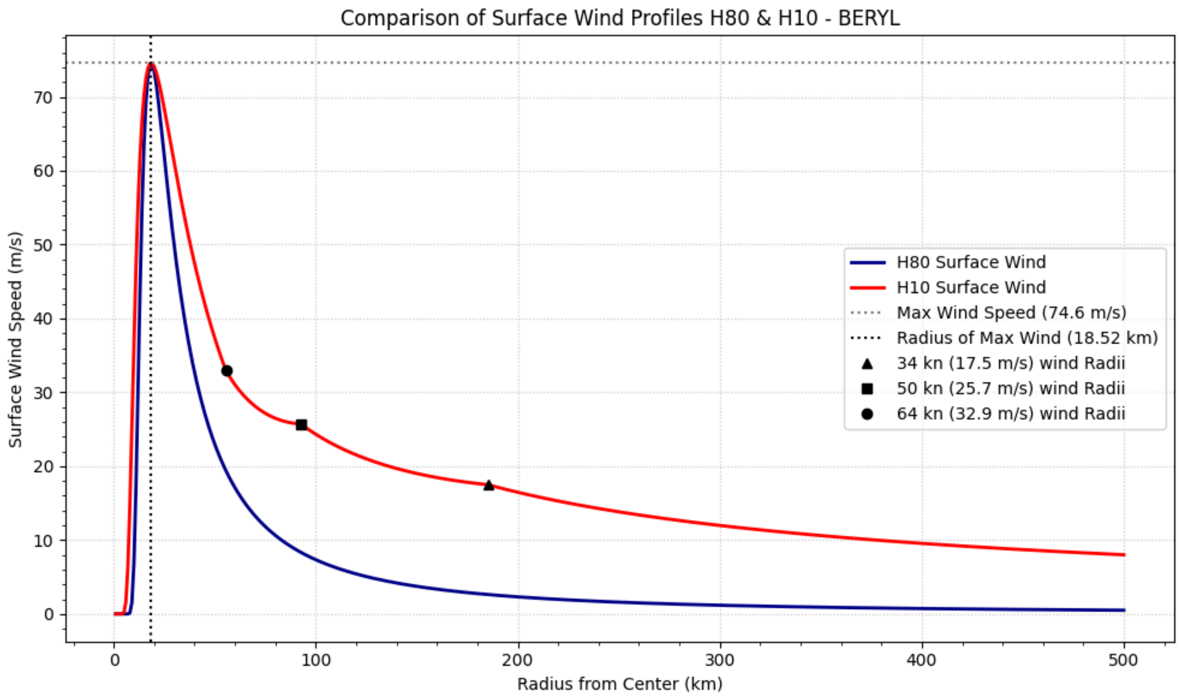
```
2.13819338789878
1.1859098300854873
```

H80 & H10 Plot

In [65]:
```
def plot_1d_profiles(r_1980_km, v_1980_s_m_s, r_2010_km, v_2010_s_m_s, hurricane_na
    RMW = float(row['RadiusOfMaxWind']) * NMI_TO_M / 1000
    plt.figure(figsize=(10, 6))

    plt.plot(r_1980_km, v_1980_s_m_s, label='H80 Surface Wind', linewidth=2, color=

    plt.plot(r_2010_km, v_2010_s_m_s, label='H10 Surface Wind', linewidth=2, color=

    v_max_m_s = v_max_kn * KNOTS_TO_M_S
    plt.axhline(v_max_m_s, color='gray', linestyle=':', label=f'Max Wind Speed ({v_

    plt.axvline(RMW, color='black', linestyle=':', label=f'Radius of Max Wind ({RMW
    plt.plot(rad34_v_m/1000, 34*KNOTS_TO_M_S, 'k^', label=f'34 kn ({34*KNOTS_TO_M_S
    plt.plot(rad50_v_m/1000, 50*KNOTS_TO_M_S, 'ks', label=f'50 kn ({50*KNOTS_TO_M_S
    plt.plot(rad64_v_m/1000, 64*KNOTS_TO_M_S, 'ko', label=f'64 kn ({64*KNOTS_TO_M_S
```

```
    plt.title(f'Comparison of Surface Wind Profiles H80 & H10 - {hurricane_name}')
    plt.xlabel('Radius from Center (km)')
    plt.ylabel('Surface Wind Speed (m/s)')
    plt.legend()
    plt.grid(True, linestyle=':', alpha=0.6)
    plt.minorticks_on()
    plt.tight_layout()
    plt.savefig(f'holland_plots/1D_H80_H10_comp_{hurricane_name}')
    plt.show()
```

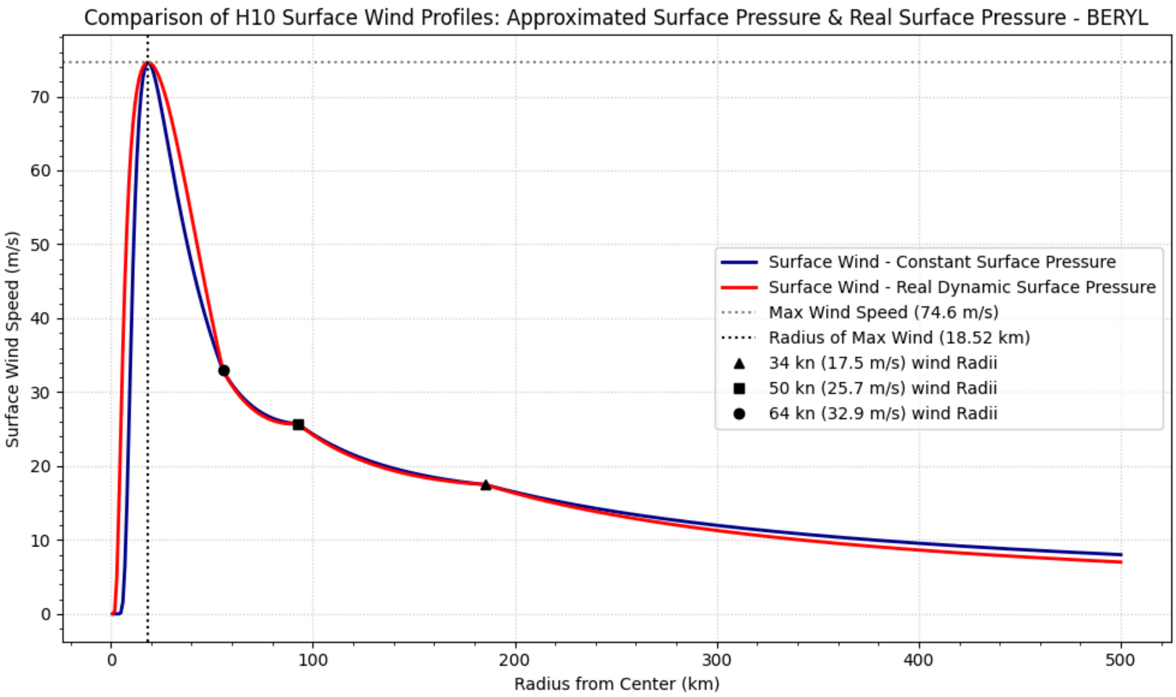
In [67]:
```
plot_1d_profiles(H80_r_km, H80_wind_ms, H10_r_km, H10_wind_ms, row['Name'], float(r
```



H10 orginal & H10 enchanced Plot

In [70]:
```
v_max_kn = float(row['MaxWind(kn)'])
hurricane_name  = row['Name']
RMW = float(row['RadiusOfMaxWind']) * NMI_TO_M / 1000
plt.figure(figsize=(10, 6))

# Plot Holland 2010 Surface Wind with constant env pressure
plt.plot(h10_df['Distance_km'], h10_df['Wind_ms'], label='Surface Wind - Constant S

# Plot Holland 2010 Surface Wind with dynamic real-time pressure
plt.plot(h10_DP_df['Distance_km'], h10_DP_df['Wind_ms'], label='Surface Wind - Real

# Vmax line (Input Surface Wind Maximum)
v_max_m_s = v_max_kn * KNOTS_TO_M_S
plt.axhline(v_max_m_s, color='gray', linestyle=':', label=f'Max Wind Speed ({v_max_

plt.axvline(RMW, color='black', linestyle=':', label=f'Radius of Max Wind ({RMW} km
```

```
plt.plot(rad34_v_m/1000, 34*KNOTS_TO_M_S, 'k^', label=f'34 kn ({34*KNOTS_TO_M_S:.1f
plt.plot(rad50_v_m/1000, 50*KNOTS_TO_M_S, 'ks', label=f'50 kn ({50*KNOTS_TO_M_S:.1f
plt.plot(rad64_v_m/1000, 64*KNOTS_TO_M_S, 'ko', label=f'64 kn ({64*KNOTS_TO_M_S:.1f

plt.title(f'Comparison of H10 Surface Wind Profiles: Approximated Surface Pressure
plt.xlabel('Radius from Center (km)')
plt.ylabel('Surface Wind Speed (m/s)')
plt.legend()
plt.grid(True, linestyle=':', alpha=0.6)
plt.minorticks_on()
plt.tight_layout()
plt.savefig(f'holland_plots/1D_H10_pressure_comp_{hurricane_name}')
plt.show()
```



Quantatitive analysis

In [74]:
```
def compare_wind_by_intervals(df1, df2, rmw):
    A = df1['Wind_ms']
    B = df2['Wind_ms']
    rmw = float(rmw)

    # Assume index = radius (km)
    radius = df1['Distance_km']

    # Define bins
    bins = [
        (0, rmw, f"0-{rmw} km"),
        (rmw, rmw+50, f"{rmw}-{rmw+50} km"),
        (rmw+50, rmw+100, f"{rmw+50}-{rmw+100} km"),
        (rmw+100, rmw+200, f"{rmw+100}-{rmw+200} km"),
        (rmw+200, rmw+300, f"{rmw+200}-{rmw+300} km"),
        (rmw+300, np.inf, f"{rmw+300}+ km")
```

```
    ]

    rows = []

    for lower, upper, label in bins:
        mask = (radius >= lower) & (radius < upper)

        if mask.sum() == 0:
            continue

        A_bin = A[mask]
        B_bin = B[mask]

        diff = A[mask] - B[mask]
        abs_diff = np.abs(diff)

        rmse = np.sqrt(np.mean(diff**2))
        mean_diff = diff.mean()
        max_dev = abs_diff.max()

        mean_ref = (A_bin.mean() + B_bin.mean()) / 2
        rmse_pct = (rmse / mean_ref) * 100 if mean_ref != 0 else np.nan

        rows.append({
            "Interval": label,
            "A Mean": A_bin.mean(),
            "B Mean": B_bin.mean(),
            "Mean difference": diff.mean(),
            "RMSE": rmse,
            "RMSE (%)": rmse_pct,
            "Max deviation": abs_diff.max(),
            "Points": mask.sum()
        })

    return pd.DataFrame(rows).set_index("Interval")
```

In [76]:

```
results = compare_wind_by_intervals(h10_df, h80_df, row['RadiusOfMaxWind'])
```

In [78]:

```
results = compare_wind_by_intervals(h10_df, h10_DP_df, row['RadiusOfMaxWind'])
```

Pressure fields data by quadrants from ERA5 notebook

In [83]:

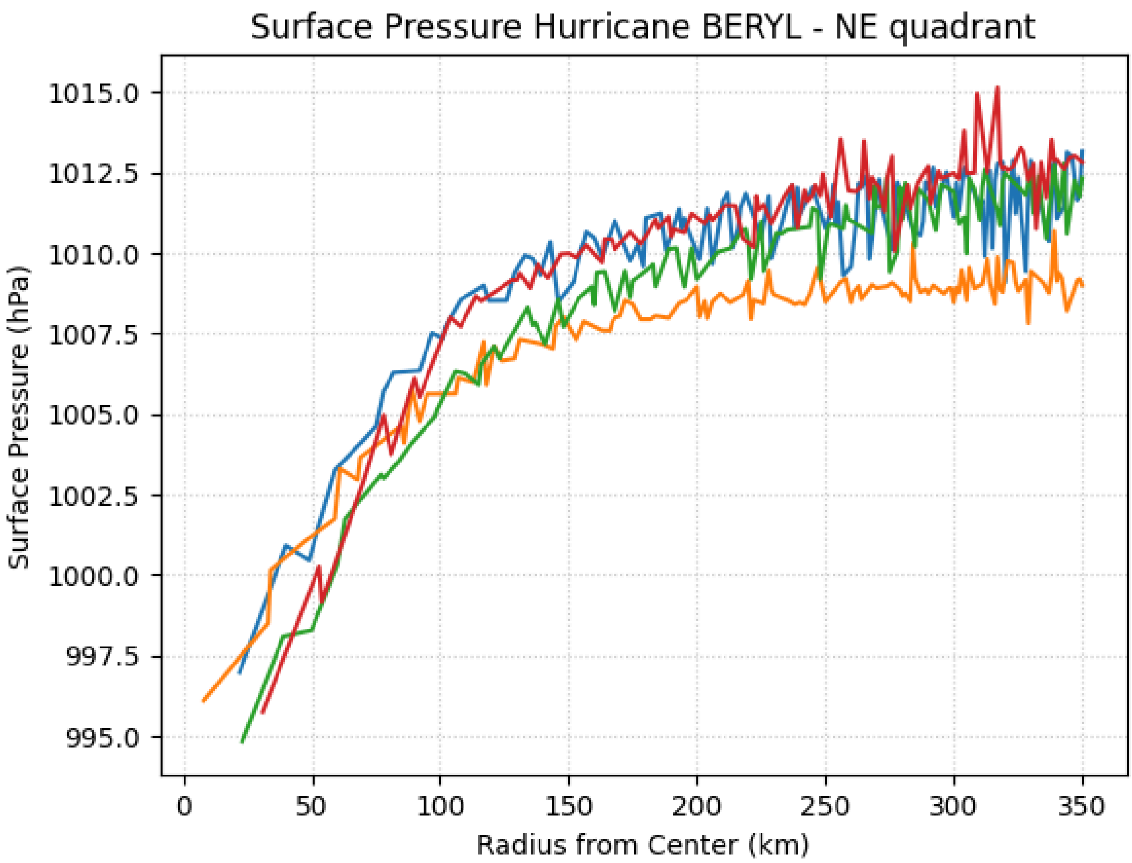
```
PN_quad = pd.read_csv("data/Beryl_pressure_average_quadrants.csv")
```

In [85]:

```
nw_df = PN_quad[['Distance', 'NW']].dropna()
ne_df = PN_quad[['Distance', 'NE']].dropna()
sw_df = PN_quad[['Distance', 'SW']].dropna()
se_df = PN_quad[['Distance', 'SE']].dropna()

r_data_NW = nw_df['Distance'].to_numpy(dtype=float) * 1000
p_data_NW = nw_df['NW'].to_numpy(dtype=float)
```

```python
r_data_NE = ne_df['Distance'].to_numpy(dtype=float) * 1000
p_data_NE = ne_df['NE'].to_numpy(dtype=float)

r_data_SW = sw_df['Distance'].to_numpy(dtype=float) * 1000
p_data_SW = sw_df['SW'].to_numpy(dtype=float)

r_data_SE = se_df['Distance'].to_numpy(dtype=float) * 1000
p_data_SE = se_df['SE'].to_numpy(dtype=float)
```

In [89]:
```python
x = np.arange(1, 351)

quadrants = ['SE', 'SW', 'NW', 'NE']

for q in quadrants:
    plt.plot(x, PN_quad[q].head(350)/100)
    plt.title(f"Surface Pressure Hurricane {row['Name']} - {q} quadrant")
    plt.xlabel('Radius from Center (km)')
    plt.ylabel('Surface Pressure (hPa)')
    plt.grid(True, linestyle=':', alpha=0.6)
    plt.savefig(f"q_pressure/Surface Pressure Hurricane {row['Name']} - {q} quadran
    #plt.show()
```



Modelling H10 enchanced by quandrants

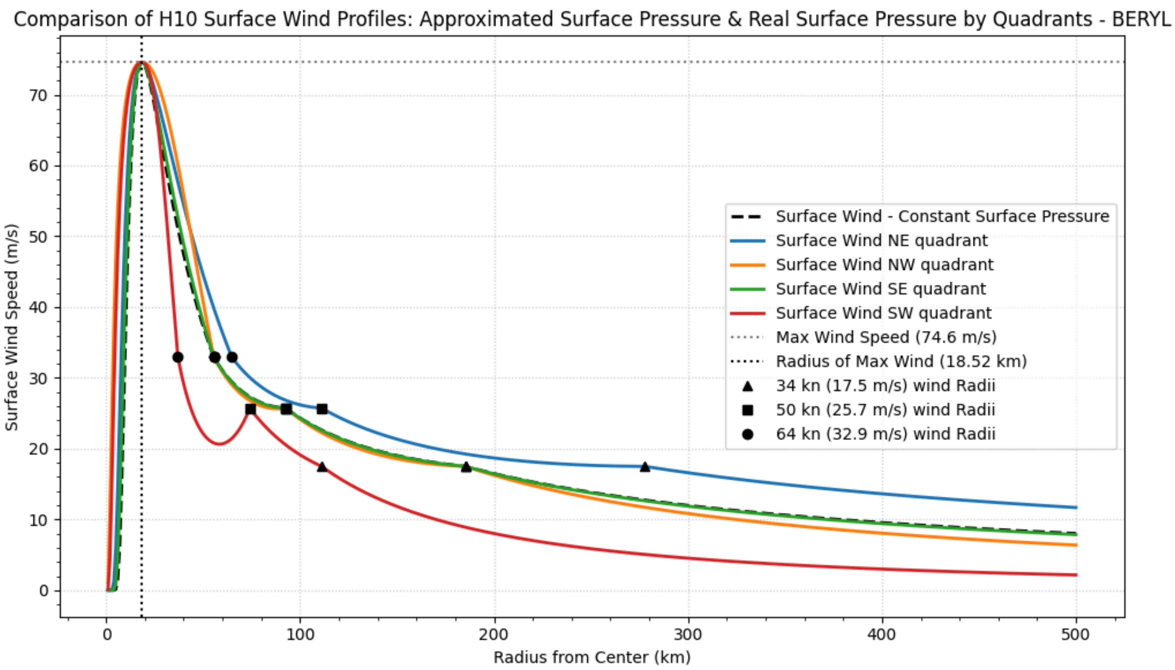
In [92]:
```
H10_r_km_NW, H10_wind_ms_NW = holland10_dynamic_pressure(r_data_NW, p_data_NW, row,
h10_NW_df = pd.DataFrame({
    'Distance_km': H10_r_km_NW,
    'Wind_ms': H10_wind_ms_NW
})

H10_r_km_NE, H10_wind_ms_NE = holland10_dynamic_pressure(r_data_NE, p_data_NE, row,
h10_NE_df = pd.DataFrame({
    'Distance_km': H10_r_km_NE,
    'Wind_ms': H10_wind_ms_NE
})

H10_r_km_SW, H10_wind_ms_SW = holland10_dynamic_pressure(r_data_SW, p_data_SW, row,
h10_SW_df = pd.DataFrame({
    'Distance_km': H10_r_km_SW,
    'Wind_ms': H10_wind_ms_SW
})

H10_r_km_SE, H10_wind_ms_SE = holland10_dynamic_pressure(r_data_SE, p_data_SE, row,
h10_SE_df = pd.DataFrame({
    'Distance_km': H10_r_km_SE,
    'Wind_ms': H10_wind_ms_SE
})
```

```
2.13819338789878
0.8483298930431606
2.13819338789878
1.6049175096576112
2.13819338789878
0.9799886654968889
2.13819338789878
1.9339987520671698
```

H10 original & H10 enchanced assymetric quandrants Plot

In [98]:
```
v_max_kn = float(row['MaxWind(kn)'])
hurricane_name  = row['Name']
RMW = float(row['RadiusOfMaxWind']) * NMI_TO_M / 1000
plt.figure(figsize=(10, 6))

# Plot Holland 2010 Surface Wind with constant env pressure
plt.plot(h10_df['Distance_km'], h10_df['Wind_ms'], label='Surface Wind - Constant S

# Plot Holland 2010 Surface Wind with dynamic real-time pressure
plt.plot(h10_NE_df['Distance_km'], h10_NE_df['Wind_ms'], label='Surface Wind NE qua
plt.plot(h10_NW_df['Distance_km'], h10_NW_df['Wind_ms'], label='Surface Wind NW qua
plt.plot(h10_SE_df['Distance_km'], h10_SE_df['Wind_ms'], label='Surface Wind SE qua
plt.plot(h10_SW_df['Distance_km'], h10_SW_df['Wind_ms'], label='Surface Wind SW qua

# Vmax line (Input Surface Wind Maximum)
v_max_m_s = v_max_kn * KNOTS_TO_M_S
plt.axhline(v_max_m_s, color='gray', linestyle=':', label=f'Max Wind Speed ({v_max_
```

```
plt.axvline(RMW, color='black', linestyle=':', label=f'Radius of Max Wind ({RMW} km

plt.plot(rad34_v_m/1000, 34*KNOTS_TO_M_S, 'k^', label=f'34 kn ({34*KNOTS_TO_M_S:.1f
plt.plot(row['34NE']*NMI_TO_KM, 34*KNOTS_TO_M_S, 'k^')
plt.plot(row['34NW']*NMI_TO_KM, 34*KNOTS_TO_M_S, 'k^')
plt.plot(row['34SE']*NMI_TO_KM, 34*KNOTS_TO_M_S, 'k^')
plt.plot(row['34SW']*NMI_TO_KM, 34*KNOTS_TO_M_S, 'k^')

plt.plot(rad50_v_m/1000, 50*KNOTS_TO_M_S, 'ks', label=f'50 kn ({50*KNOTS_TO_M_S:.1f
plt.plot(row['50NE']*NMI_TO_KM, 50*KNOTS_TO_M_S, 'ks')
plt.plot(row['50NW']*NMI_TO_KM, 50*KNOTS_TO_M_S, 'ks')
plt.plot(row['50SE']*NMI_TO_KM, 50*KNOTS_TO_M_S, 'ks')
plt.plot(row['50SW']*NMI_TO_KM, 50*KNOTS_TO_M_S, 'ks')

plt.plot(rad64_v_m/1000, 64*KNOTS_TO_M_S, 'ko', label=f'64 kn ({64*KNOTS_TO_M_S:.1f
plt.plot(row['64NE']*NMI_TO_KM, 64*KNOTS_TO_M_S, 'ko')
plt.plot(row['64NW']*NMI_TO_KM, 64*KNOTS_TO_M_S, 'ko')
plt.plot(row['64SE']*NMI_TO_KM, 64*KNOTS_TO_M_S, 'ko')
plt.plot(row['64SW']*NMI_TO_KM, 64*KNOTS_TO_M_S, 'ko')

plt.title(f'Comparison of H10 Surface Wind Profiles: Approximated Surface Pressure
plt.xlabel('Radius from Center (km)')
plt.ylabel('Surface Wind Speed (m/s)')
plt.legend()
plt.grid(True, linestyle=':', alpha=0.6)
plt.minorticks_on()
plt.tight_layout()
plt.savefig(f'holland_plots/1D_H10_pressure_comp_quads_{hurricane_name}')
plt.show()
```



Appendix B

Jupyter Notebook ERA5 Pressure Processing

Python Packages used

In [6]:
```
import cdsapi
import pandas as pd
import xarray as xr
import numpy as np
import matplotlib.pyplot as plt
from netCDF4 import Dataset
```

API request for ERA5 reanalysis data

In [10]:
```
c = cdsapi.Client()
print("CDS API authentication successful")
```

```
CDS API authentication successful
```

In [ ]:
```
dataset = "reanalysis-era5-single-levels"
request = {
    "product_type": ["reanalysis"],
    "variable": ["mean_sea_level_pressure"],
    "year": ["2024"],
    "month": [
        "06", "07", "09",
        "10"
    ],
    "day": [
        "01", "02", "03",
        "04", "05", "06",
        "07", "08", "09",
        "10", "11", "12",
        "13", "14", "15",
        "16", "17", "18",
        "19", "20", "21",
        "22", "23", "24",
        "25", "26", "27",
        "28", "29", "30",
        "31"
    ],
    "time": [
        "00:00", "01:00", "02:00",
        "03:00", "04:00", "05:00",
        "06:00", "07:00", "08:00",
        "09:00", "10:00", "11:00",
        "12:00", "13:00", "14:00",
        "15:00", "16:00", "17:00",
        "18:00", "19:00", "20:00",
        "21:00", "22:00", "23:00"
    ],
    "data_format": "netcdf",
    "download_format": "unarchived",
```

```
    "area": [50, 100, 0, 0]
}

client = cdsapi.Client()
client.retrieve(dataset, request).download()
```

```
2026-04-24 20:31:42,384 INFO [2025-12-11T00:00:00] Please note that a dedicated cata
logue entry for this dataset, post-processed and stored in Analysis Ready Cloud Opti
mized (ARCO) format (Zarr), is available for optimised time-series retrievals (i.e.
for retrieving data from selected variables for a single point over an extended peri
od of time in an efficient way). You can discover it [here](https://cds.climate.cope
rnicus.eu/datasets/reanalysis-era5-single-levels-timeseries?tab=overview)
2026-04-24 20:31:42,385 INFO Request ID is 6dd22222-2e4b-4c4e-8e02-67d048f06615
2026-04-24 20:31:42,480 INFO status has been updated to accepted
```

In [566…

```
from netCDF4 import Dataset
Dataset("e1d7810ecd578995851fb2b581ec9581.nc")
```

Out[566…

```
<class 'netCDF4.Dataset'>
root group (NETCDF4 data model, file format HDF5):
    GRIB_centre: ecmf
    GRIB_centreDescription: European Centre for Medium-Range Weather Forecasts
    GRIB_subCentre: 0
    Conventions: CF-1.7
    institution: European Centre for Medium-Range Weather Forecasts
    history: 2026-02-25T08:03 GRIB to CDM+CF via cfgrib-0.9.15.1/ecCodes-2.42.0 wi
th {"source": "tmp_yzfn4cr/data.grib", "filter_by_keys": {"stream": ["oper"], "ste
pType": ["instant"]}, "encode_cf": ["parameter", "time", "geography", "vertical"]}
    dimensions(sizes): valid_time(2928), latitude(201), longitude(401)
    variables(dimensions): int64 number(), int64 valid_time(valid_time), float64 l
atitude(latitude), float64 longitude(longitude), <class 'str'> expver(valid_time),
float32 sp(valid_time, latitude, longitude)
    groups:
```

In [ ]:

```
ds = xr.open_dataset("e1d7810ecd578995851fb2b581ec9581.nc")
```

Data Processing

In [109…

```
def get_local_sp(ds, center_lat, center_lon, radius=3):
    lat_min = center_lat - radius
    lat_max = center_lat + radius
    lon_min = center_lon - radius
    lon_max = center_lon + radius

    # Slice the dataset
    sp_local = ds['sp'].sel(
        latitude=slice(lat_max, lat_min),
        longitude=slice(lon_min, lon_max)
    )

    return sp_local
```

In [109…

```
hurricane_peaks = pd.DataFrame({
    'name': ['Helene', 'Oscar', 'Kirk', 'Beryl'],
```

```
    'date': ['2024-09-27', '2024-10-20', '2024-10-04', '2024-07-02'],
    'time': ['00:00', '22:00', '06:00', '09:45'],  # in UTC
    'lat': [28.7, 20.3, 21.9, 14.8],
    'lon': [-84.3, -74.4, -47.9, -67.2]
})

hurricane_peaks['datetime'] = pd.to_datetime(hurricane_peaks['date'] + ' ' + hurric
```

In [ ]:
```
oscar = hurricane_peaks[1]
```

In [ ]:
```
sp_local = get_local_sp(ds.sel(valid_time=oscar['datetime'], method='nearest'),
                        oscar['lat'], oscar['lon'], radius=3)
sp_local_hPa = sp_local / 100.0
```

Haversine Formula

In [109…
```
R = 6371.0

lat0 = oscar['lat']
lon0 = oscar['lon']

lat = np.deg2rad(sp_local.latitude.values)
lon = np.deg2rad(sp_local.longitude.values)
lat0 = np.deg2rad(lat0)
lon0 = np.deg2rad(lon0)

lon2d, lat2d = np.meshgrid(lon, lat)

dlat = lat2d - lat0
dlon = lon2d - lon0

a = np.sin(dlat/2)**2 + np.cos(lat0) * np.cos(lat2d) * np.sin(dlon/2)**2
c = 2 * np.arcsin(np.sqrt(a))

distance_km = R * c
```

In [113…
```
dist_flat = distance_km.flatten()
pressure_flat = sp_local_hPa.values.flatten()
bins = np.arange(0, np.max(dist_flat) + 1, 1)
bin_indices = np.digitize(dist_flat, bins)
```

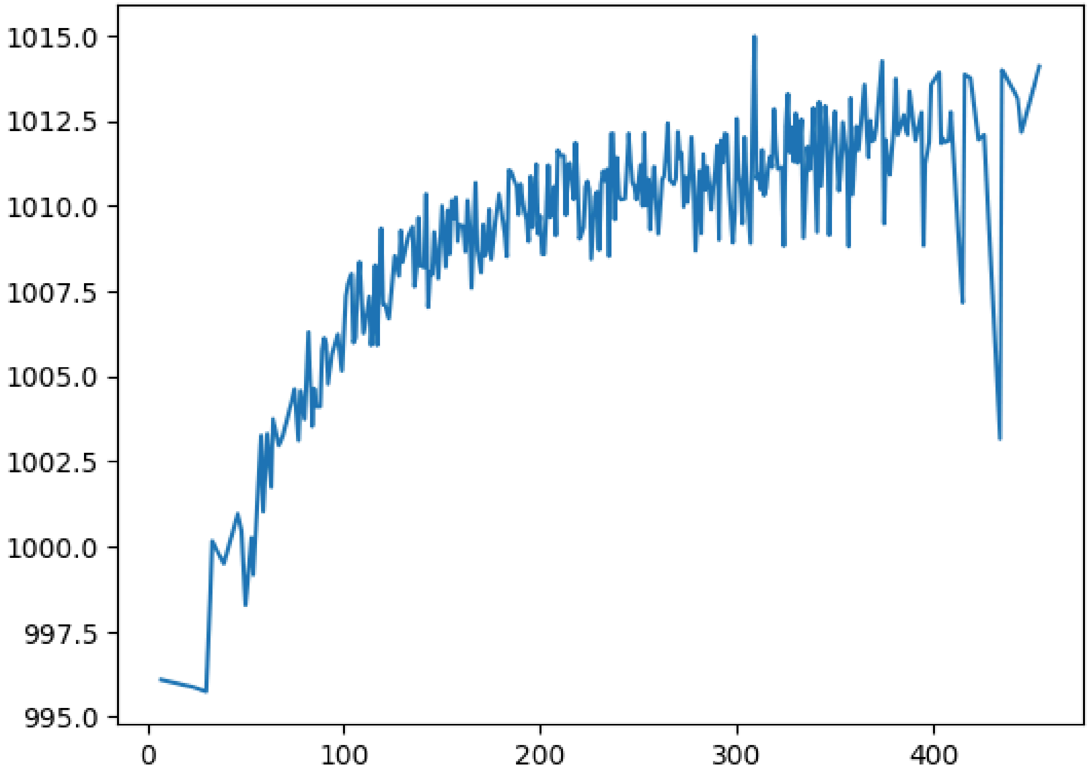
In [ ]:
```
mean_pressure = np.array([
    pressure_flat[bin_indices == i].mean()
    for i in range(1, len(bins))
])
```

In [110…
```
mean_pressure = pd.Series(mean_pressure).interpolate().values
```

In [110…
```
mean_pressure_df = pd.DataFrame(mean_pressure)
mean_pressure_df.columns = ["SP"]
mean_pressure_df.to_csv(f"{oscar['name']}_pressure_average.csv")
```

In [110…

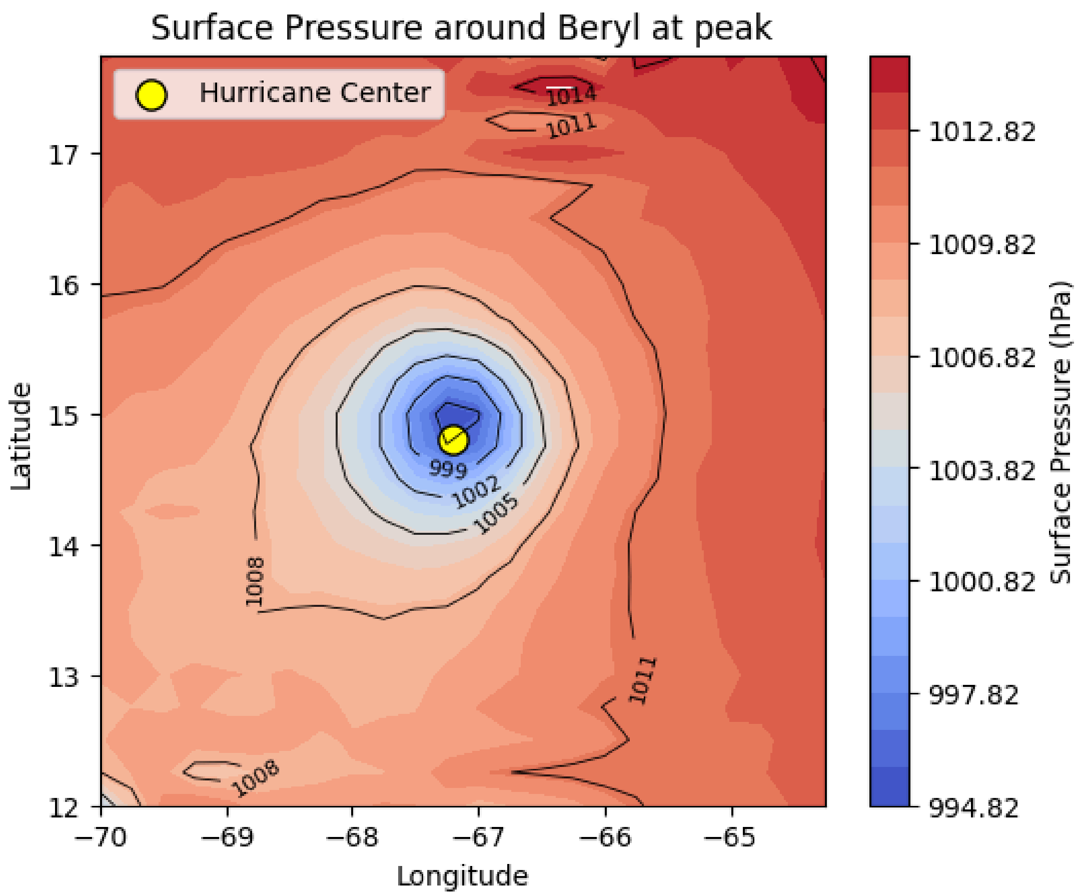
```
x = np.arange(0,mean_pressure.size)
plt.plot(x, mean_pressure)
plt.show()
```



Pressure field Plot

In [110…

```
plt.figure(figsize=(6,5))
levels = np.arange(sp_local_hPa.min(), sp_local_hPa.max(), 1)
cf = plt.contourf(sp_local_hPa.longitude, sp_local_hPa.latitude, sp_local_hPa, leve
cbar = plt.colorbar(cf)
cbar.set_label('Surface Pressure (hPa)')

cs = plt.contour(sp_local_hPa.longitude, sp_local_hPa.latitude, sp_local_hPa, color

plt.clabel(cs, fontsize=8, inline=True)

plt.scatter(oscar['lon'], oscar['lat'], color='yellow', edgecolor='black', s=120, m

plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.title(f"Surface Pressure around {oscar['name']} at peak")
plt.legend()
plt.savefig(f"pressure_plots/{oscar['name']}_pressure_around")
plt.show()
```



Computing radial pressure mean profiles in NE/NW/SE/SW quadrants

In [111…

```
import numpy as np

def quadrant_radial_profiles(sp_local, center_lat, center_lon, r_max=800, dr=10):
    R = 6371.0

    lat_vals = sp_local.latitude.values
    lon_vals = sp_local.longitude.values

    lat = np.deg2rad(lat_vals)
    lon = np.deg2rad(lon_vals)
    lat0 = np.deg2rad(center_lat)
    lon0 = np.deg2rad(center_lon)

    lon2d, lat2d = np.meshgrid(lon, lat)

    dlat = lat2d - lat0
    dlon = lon2d - lon0

    a = np.sin(dlat / 2.0) ** 2 + np.cos(lat0) * np.cos(lat2d) * np.sin(dlon / 2.0)
    c = 2.0 * np.arcsin(np.sqrt(a))
    distance_km = R * c
```

```
    # quadrant masks in degrees
    lon_deg2d, lat_deg2d = np.meshgrid(lon_vals, lat_vals)
    dlat_deg = lat_deg2d - center_lat
    dlon_deg = lon_deg2d - center_lon

    mask_NE = (dlat_deg >= 0) & (dlon_deg >= 0)
    mask_NW = (dlat_deg >= 0) & (dlon_deg < 0)
    mask_SE = (dlat_deg < 0) & (dlon_deg >= 0)
    mask_SW = (dlat_deg < 0) & (dlon_deg < 0)

    quadrant_masks = {
        'NE': mask_NE,
        'NW': mask_NW,
        'SE': mask_SE,
        'SW': mask_SW
    }

    field = sp_local.values

    r_km = np.arange(1, r_max + 1)

    profiles = {q: np.full(len(r_km), np.nan) for q in quadrant_masks}

    for q, qmask in quadrant_masks.items():
        for i, r in enumerate(r_km):
            r1 = r - 0.5
            r2 = r + 0.5

            radial_mask = (distance_km >= r1) & (distance_km < r2)
            mask = radial_mask & qmask

            vals = field[mask]
            vals = vals[~np.isnan(vals)]

            if len(vals) > 0:
                profiles[q][i] = np.mean(vals)

    return r_km, profiles
```

In [107…
```
r_centers, p_quads = quadrant_radial_profiles(
    sp_local,
    center_lat=oscar['lat'],
    center_lon=oscar['lon'],
    r_max=600,
    dr=1
)
```

In [107…
```
p_quads_interp = {
    q: pd.Series(p_quads[q]).interpolate(limit_area='inside').values
    for q in p_quads
}
```

In [113…
```
p_quads_interp_df = pd.DataFrame(p_quads_interp)
p_quads_interp_df['Distance'] = r_centers
```

```
p_quads_interp_df.to_csv(f"{oscar['name']}_pressure_average_quadrants.csv", index=F
```